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Physics of Living Systems

# Quantifying the shape of cells - from Minkowski tensors to p-atic orders

**Lea Happel, Griseldis Oberschelp, Valeriia Grudtsyna, Harish P Jain, Rastko Sknepnek, Amin Doostmohammadi, Axel Voigt**

Institute of Scientific Computing, TU Dresden, Dresden, Germany • Niels Bohr Institute, University of Copenhagen, Kobenhavn, Denmark • Njord Centre, Physics Department, University of Oslo, Oslo, Norway • School of Life Sciences, University of Dundee, Dundee, United Kingdom • School of Science and Engineering, University of Dundee, Dundee, United Kingdom • Center of Systems Biology Dresden, Dresden, Germany • Cluster of Excellence, Physics of Life, TU Dresden, Dresden, Germany



**eLife Assessment**

This **important** work describes a set of parameters that give a robust description of shape features of cells in tissues. The evidence for the usefulness of these parameters is **solid**. The work should be of interest for anybody analyzing epithelial dynamics, but more details about the analysis of experimental images are necessary and some streamlining of the text would increase the accessibility of the material for non-specialists.

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## Abstract

*P*-atic liquid crystal theories offer new perspectives on how cells self-organize and respond to mechanical cues. Understanding and quantifying the underlying orientational orders is therefore essential for unraveling the physical mechanisms that govern tissue dynamics. Due to the deformability of cells this requires quantifying their shape. We introduce rigorous mathematical tools and a reliable framework for such shape analysis. Applying this to segmented cells in MDCK monolayers and computational approaches for active vertex models and multiphase field models allows to demonstrate independence of shape measures and the presence of various *p*-atic orders at the same time. This challenges previous findings and opens new pathways for understanding the role of orientational symmetries and *p*-atic liquid crystal theories in tissue mechanics and development.

## Introduction

The importance of orientational order in biological systems is becoming increasingly clear, as it plays a critical role in processes such as tissue morphogenesis, collective cell motion, and cellular extrusion. Orientational order results from the shapes of cells and their alignments with neighboring cells. Disruptions in this order, known as topological defects, are often linked to key biological events. For instance, defects—points or lines where the order breaks down—can drive cell extrusion (Saw et al., 2017; Monfared et al., 2023) or trigger morphological changes in tissues (Maroudas-Sacks et al., 2021; Ravichandran et al., 2024). Orientational order is linked to liquid crystal theories (de Gennes and Prost, 1993) and should here be interpreted in a broad sense. Recent evidence extends beyond nematic order, characterized by symmetry under $180^\circ = 2\pi/2$ rotation, to higher-order symmetries such as tetratic order ($90^\circ = 2\pi/4$), hexatic order ($60^\circ = 2\pi/6$), and even general $p$-atic orders ($2\pi/p$, $p$ being an integer). In biological contexts, nematic order ($p = 2$) has been widely studied in epithelial tissues (Duclos et al., 2017; Saw et al., 2017; Kawaguchi et al., 2017), linking defects to cellular behaviors and tissue organization. More complex orders, such as tetratic order ($p = 4$) (Cislo et al., 2023) and hexatic oder ($p = 6$) (Li and Ciamarra, 2018; Durand and Heu, 2019; Pasupalak et al., 2020; Armengol-Collado et al., 2023; Eckert et al., 2023), have also been observed in experimental systems, offering new perspectives on how cells self-organize and respond to mechanical cues. Understanding and quantifying orientational order and the corresponding liquid crystal theory are therefore essential for unraveling the physical mechanisms that govern biological dynamics.

In physics, $p$-atic liquid crystals illustrate how particle shape and symmetry influence phase behavior, with seminal works dating back to Onsager's theories (Onsager, 1949). Most prominently, hexatic order ($p = 6$) has been postulated and found by experiments and simulations as an intermediate state between crystalline solid and isotropic liquid in (Halperin and Nelson, 1978; Nelson and Halperin, 1979; ***Murray and Van Winkle, 1987***; Bladon and Frenkel, 1995; Zahn et al., 1999; Gasser et al., 2010; Bernard and Krauth, 2011). Other examples are colloidal systems for triadic platelets (Bowick et al., 2017) or cubes (Wojciechowski and Frenkel, 2004), which lead to $p$-atic order with $p = 3$ and $p = 4$, respectively. Even pentatic ($p = 5$) and heptatic ($p = 7$) liquid crystals have been engineered (Wang and Mason, 2018; Yu and Mason, 2023). The corresponding liquid crystal theories for $p$-atic order have only recently been defined (Giomi et al., 2022a; Krommydas et al., 2023) and can also be used in biological contexts. However, while in colloidal systems particle shapes remain fixed, in biological tissues, cells are dynamic: their shapes are irregular, variable, and influenced by internal and external forces. These unique properties make quantifying $p$-atic order in tissues significantly more challenging, as quantification of the cell shapes is required. We will demonstrate that existing methods, such as bond-orientational order (Nelson and Halperin, 1979) or polygonal shape analysis (Armengol-Collado et al., 2023), might fail to capture the nuances of irregular cell shapes, which has severe consequences on the definition of $p$-atic order.

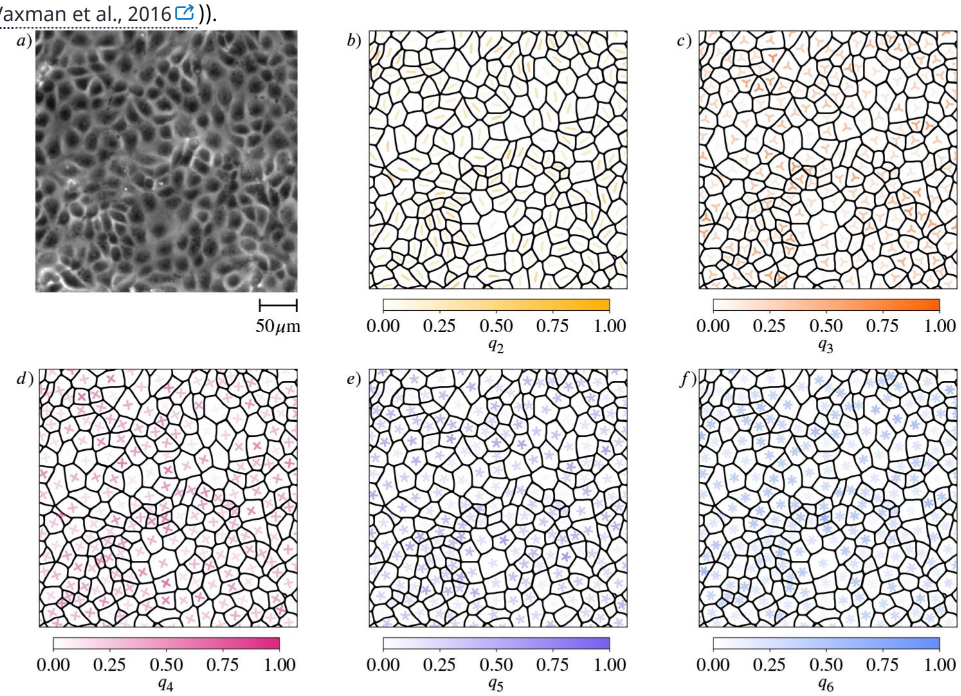
To address this, we adapt Minkowski tensors (Mecke, 2000; Schröder-Turk et al., 2013)—rigorous mathematical tools for shape analysis—to quantify $p$-atic orders in cell monolayers. Minkowski tensors provide robust and sensitive measures of shape anisotropy and orientation, accommodating both smooth and polygonal shapes while remaining resilient to small perturbations. By applying these tools, we re-examine previous conclusions about $p$-atic orders in epithelial tissues and demonstrate that certain widely accepted results require reconsideration.

Our study leverages a combination of experimental and computational approaches. Experimentally, we analyze confluent monolayers of MDCK (Madin-Darby Canine Kidney) cells. **Figure 1** provide a snapshot of a considered monolayer of 235 wild-type MDCK cells, together with their shape classification by Minkowski tensors. The corresponding statistical data and probability distributions of these quantities are shown in **Figure 2**. These data indicate the presence of all $p$-atic orders at once with similar probability distributions, mean values and standard devistions. In addition to these experiments we also analyze data of MDCK cells reported in (Armengol-Collado et al., 2023). Computationally, we employ two complementary models for cell monolayers: the active vertex model and the multiphase field model. These approaches allow us to systematically vary parameters such as cell activity and mechanical properties, providing a comprehensive view of how $p$-atic orders emerges in different contexts.

## Figure 1.

### Shape classification of cells in wild-type MDCK cell monolayer.

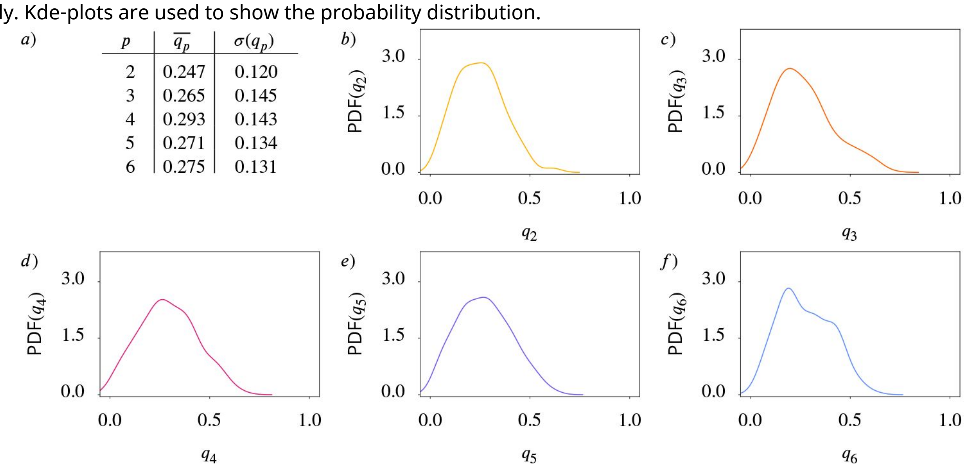
a) Raw experimental data. b) - f) Minkowski tensor, visualized using $\vartheta_p$ and $q_p$, Equation 7 (see Methods and materials) for $p$ = 2, 3, 4, 5, 6, respectively. The brightness and the rotation of the $p$-atic director indicates the magnitude and the orientation, respectively. The visualization uses rotationally-symmetric direction fields (known as $p$-RoSy fields in computer graphics (Vaxman et al., 2016)).




## Figure 2.

### Statistical data for cell shapes identified in Figure 1 (see Methods and materials).

a) Mean $\overline{q_p}$ and standard deviation $\sigma(q_p)$ of $q_p$. b) - f) Probability distribution function (PDF) of $q_p$ for $p$ = 2, 3, 4, 5, 6, respectively. Kde-plots are used to show the probability distribution.




| $p$ | $\overline{q_p}$ | $\sigma(q_p)$ |
|---|---|---|
| 2 | 0.247 | 0.120 |
| 3 | 0.265 | 0.145 |
| 4 | 0.293 | 0.143 |
| 5 | 0.271 | 0.134 |
| 6 | 0.275 | 0.131 |

By combining these robust mathematical tools, experimental insights, and computational models, we establish a reliable framework for quantifying $p$-atic orders in biological tissues. This framework not only challenges previous findings, e.g. a proposed hexatic-nematic crossover on larger length scales (Armengol-Collado et al., 2023), but also opens new pathways for understanding the role of orientational symmetries in tissue mechanics and development, which require to consider multiple orientational symmetries.

# Methods and materials

## Shape classification

### Minkowski tensors

Essential properties of the geometry of a two-dimensional object are summarized by scalar-valued size measures, so called Minkowski functionals. They are defined by

$$W_0(C) = \int_C \mathrm{d}C, \quad W_1(C) = \int_{\partial C} \mathrm{d}\partial C, \text{ and } \quad W_2(C) = \int_{\partial C} \mathcal{H}\, \mathrm{d}\partial C, \tag{1}$$

where $\mathscr{C}$ is the smooth two-dimensional object, with contour $\partial\mathscr{C}$ and $\mathscr{H}$ denotes the curvature of the contour $\partial\mathscr{C}$ (see **Figure 3** for a schematic description), as reviewed in Mecke (2000). They describe the area, the perimeter and a curvature-weighted integral of the contour, respectively. Minkowski functionals are also known as intrinsic volumes. For convex objects, they have been shown to be continuous and invariant to translations and rotations. Due to these properties and Hadwiger's characterization theorem (Hadwiger, 1975), they are natural size descriptors that provide essential and complete information about invariant geometric features. However, these properties also set limits to their use as shape descriptors. Due to rotation invariance, they are unable to capture the orientation of the shape. To describe more complex shape information the scalar-valued Minkowski functionals are extended to a set of tensor-valued descriptors, known as Minkowski tensors. These objects have been investigated both in mathematical (Alesker, 1999; Hug et al., 2008, ***2007***; McMullen, 1997) and in physical literature (Schröder-Turk et al., 2010a; Kapfer et al., 2012). Using tensor products of the position vectors $\mathbf{r}$ and the normal vectors $\mathbf{n}$ of the contour $\partial\mathscr{C}$, defined as $\mathbf{r}^a \odot \mathbf{n}^b = \mathbf{r} \odot \ldots \odot \mathbf{r} \odot \mathbf{n} \odot \ldots \odot \mathbf{n}$, the first considered $a$ times and the last $b$ times with $\odot$ denoting the symmetrized tensor product, the Minkowski tensors are defined as

$$\mathbf{W}_0^{a,0}(C) = \int_C \mathbf{r}^a \, \mathrm{d}C, \quad \mathbf{W}_1^{a,b}(C) = \int_{\partial C} \mathbf{r}^a \odot \mathbf{n}^b \, \mathrm{d}\partial C, \text{ and } \quad \mathbf{W}_2^{a,b}(C) = \int_{\partial C} \mathbf{r}^a \odot \mathbf{n}^b \mathcal{H}\, \mathrm{d}\partial C. \tag{2}$$

For $\mathbf{W}_j^{0,0}(C) = W_j(C)$ for $j = 0, 1, 2$, so the Minkowski tensors are extensions of the Minkowski functionals. In analogy to Minkowski functionals also Minkowski tensors have been shown to be continuous for convex objects. Also, analogies to Hadwiger's characterization theorem exist (Alesker, 1999). However, for our purpose, the essential property is the continuity. It makes Minkowski functionals and Minkowski tensors a robust measure as small shape changes lead to small changes in the shape descriptor. Due to this property, Minkowski tensors have been successfully used as shape descriptors in different fields, e.g. in materials science as a robust measure of anisotropy in porous (Schröder-Turk et al., 2010b) and granular material (Schröder-Turk et al., 2013), in astrophysics to describe morphology of galaxies (Beisbart et al., 2002), and in biology to distinguish shapes of different types of neuronal cell networks (Beisbart et al., 2006) and to determine the direction of elongation in multiphase field models for epithelial tissue (Mueller et al., 2019; Wenzel and Voigt, 2021; Happel and Voigt, 2024). However, the mentioned examples exclusively use lower-rank Minkowski tensors $a + b \leq 2$. Higher rank tensors have not been considered in such applications but the theory also guarantees robust description of

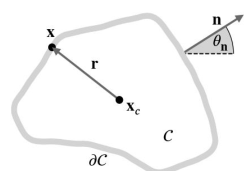



**Figure 3.**

**Schematic description of a two-dimensional object $\mathscr{C}$ with contour $\partial\mathscr{C}$.**

We denote the center of mass with $\mathbf{x}_c$ and vectors from $\mathbf{x}_c$ to points $\mathbf{x}$ on $\partial\mathscr{C}$ with $\mathbf{r}$. The outward-pointing normals are denoted by $\mathbf{n}$, the corresponding angle with the $x$-axis by $\theta_{\mathbf{n}}$.

$p$-atic orientation for $p = 3, 4, 5, 6, \ldots$. These results hold for smooth contours but also can be extended to polyconvex shapes. Furthermore, known counter-examples for non-convex shapes have very little to no relevance in applications (Kapfer, 2012).

Even if all Minkowski tensors carry important geometric information, one type is particularly interesting and will be considered in the following:

$$\boldsymbol{W}_1^{0,p}(C) = \int_{\partial C} \underbrace{\mathbf{n} \odot \mathbf{n} \odot \ldots \odot \mathbf{n}}_{\text{p}-times} \, \mathrm{d}\partial C. \quad (3)$$

In this case, the symmetrized tensor product agrees with the classical tensor product. If applied to polygonal shapes the Minkowski tensors $\boldsymbol{W}_1^{0,p}(C)$ are related to the Minkowski problem for convex polytops (i.e., generalizations of three-dimensional polyhedra to an arbitrary number of dimensions), which states that convex polytops are uniquely described by the outer normals of the edges and the length of the corresponding edge (Minkowski, 1897). Several generalizations of this result exist (Klain, 2004; Schneider, 2013), which makes the normal vectors a preferable quantity to describe shapes. However, Minkowski tensors contain redundant information, which asks for an irreducible representation. This can be achieved by decomposing the tensor with respect to the rotation group $SO(2)$ (Kapfer, 2012; Mickel et al., 2013; Klatt et al., 2022). Following this approach, one can write

$$\boldsymbol{W}_1^{0,p}(C) = \int_{S^1} \mathbf{u}^p \Psi_C(\mathbf{u}) \, \mathrm{d}\mathbf{u} \quad \text{with} \quad \Psi_C(\mathbf{u}) = \int_{\partial C} \delta(\mathbf{n}(\mathbf{x}) - \mathbf{u}) \, \mathrm{d}\partial C, \quad (4)$$

with $\Psi_{\mathscr{C}}(\mathbf{u})$ being proportional to the probability density of the normal vectors. Identifying $\mathbf{u} \in S^1$ by the angle $\theta$ between $\mathbf{u}$ and the $x$−axis allows to write

$$\Psi_C(\theta) = \sum_{p=-\infty}^{\infty} \psi_p(C) e^{-ip\theta} \quad \text{with} \quad \psi_p(C) = \frac{1}{2\pi} \int_0^{2\pi} \Psi_C(\theta) e^{ip\theta} \, \mathrm{d}\theta. \quad (5)$$

The Fourier coefficients $\psi_p(\mathscr{C})$ are the irreducible representations of $\boldsymbol{W}_1^{0,p}(C)$ and can be written as

$$\psi_p(C) = \frac{1}{2\pi} \int_{\partial C} e^{ip\theta_n} \, \mathrm{d}\partial C, \quad (6)$$

where $\theta_{\boldsymbol{n}}$ is the orientation of the outward pointing normal $\mathbf{n}$. The phase of the complex number $\psi_p(\mathscr{C})$ contains information about the preferred orientation and the absolute value $|\psi_p(\mathscr{C})|$ is a scalar index. We thus define

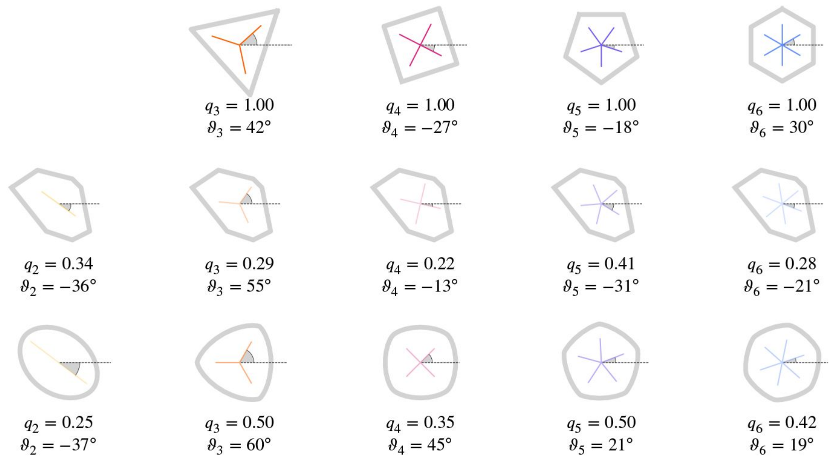
$$\vartheta_p(C) = \frac{1}{p} \arctan 2\left(\Im\psi_p(C), \Re\psi_p(C)\right) + \frac{\pi}{p} \quad \text{and} \quad q_p(C) = \frac{|\psi_p(C)|}{\psi_0(C)}, \quad (7)$$

with $\Im\psi_p$ and $\Re\psi_p$ the imaginary and real part of $\psi_p$, respectively, and $\psi_0(C) = \frac{W_1(C)}{2\pi}$. We have $q_p(\mathscr{C}) \in [0, 1]$ quantifying the strength of $p$-atic order. **Figure 4** illustrates the concept for polygonal and smooth shapes and **Figure 5** illustrates the calculation of $q_p$ and $\vartheta_p$ at the example of an equilateral triangle. Note that the constant factor of $1/(2\pi)$ in **Equation 6** does not influence the result of **Equation 7** and is therefore disregarded in **Figure 5**. Alternative derivations of **Equation 7** consider higher order trace-less tensors (Virga, 2015; Giomi et al., 2022a; Armengol-Collado et al., 2023) generated by the normal $\mathbf{n}$. In this case $\vartheta_p(\mathscr{C})$ (orange triatic director in **Figure 5** right) correspond to the eigenvector to the negative eigenvalue and $\vartheta_p(\mathscr{C}) - \pi/p$ (gray triatic director in **Figure 5** right) to the eigenvector to the positive eigenvalue. For $p = 2$ this tensor based approach corresponds to the structure tensor (Mueller et al., 2019).

## Figure 4.

### Regular and irregular shapes, adapted from Armengol-Collado et al. (2023) and Schaller et al. (2024), with magnitude and orientation calculated by Equation 7.

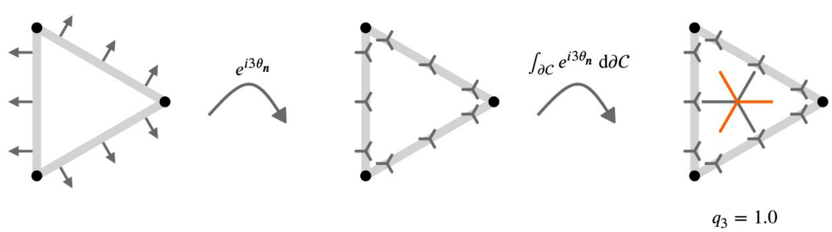
For regular shapes, the corresponding magnitude of $q_p$ is always 1.0 and the detected angle is the minimal angle of the *p*-atic orientation with respect to the *x*-axis. Note that no shape with $q_2$ = 1.0 is shown, as this would be a line. The visualization is according to **Figure 1**.




## Figure 5.

### Illustrative description of the definition of $q_3$ for an equilateral triangle.

Considering rotational symmetries under a rotation 120° = $2\pi/3$ means that vectors with an angle of 120° or 240° are treated as equal. Applied to the normals **n** (left), this means that under this rotational symmetry the normals on the three different edges are equal. Mathematically this is expressed through $e^{3i\theta_\mathbf{n}}$ resulting in the triatic director shown instead of the normals **n** (middle). One leg of the triatic director always points in the direction as the normal. While only shown for 3 points on each edge, we obtain an orientation with the respective symmetry on every point of the contour $\partial\mathscr{C}$. Considering the line integral along the contour provides the dominating triatic director, shown in the center of mass (right). To get a value between 0.0 and 1.0 for $q_3$ we normalize this integral with the length of the contour, which corresponds to $q_0$. As all triatic directors point in the same direction we obtain $q_3$ = 1.0 in this specific example. To be consistent with other approaches we rotate the resulting triatic director by 60° = $\pi/3$ leading to the orange triadic director, which is the quantity used for visualization.



## Alternative shape measures

As pointed out in the introduction, $p$-atic orders have already been identified in biological tissue and model systems. However, the way to quantify rotational symmetry in these works differs from the Minkowski tensors. In Armengol-Collado et al. (2023) the cell contour is approximated by a polygon with $V$ vertices having coordinates $\mathbf{x}_i$. The considered polygonal shape analysis is based on the shape function

$$\gamma_p(C) = \frac{\sum_{i=1}^{V} |\mathbf{r}_i|^p e^{ip\theta_i}}{\sum_{i=1}^{V} |\mathbf{r}_i|^p}, \tag{8}$$

with $\mathbf{r}_i = \mathbf{x}_i - \mathbf{x}_c$ being the vector from the centroid $\mathbf{x}_c$ and $\theta_i$ being the orientation of the $i$-th vertex of the polygon with respect to its center of mass. As the centroid is computed by $\mathbf{x}_c = \frac{1}{V}\sum_{i=1}^{V} \mathbf{x}_i$ Armengol-Collado et al. (2023) the only input data are the coordinates of the vertices of the $V$-sided polygon. Equation 8 captures the degree of regularity of the polygon by its amplitude $0 \leq |\gamma_p| \leq 1$ and its orientation $\vartheta_p^{\gamma}$ which follows as

$$\vartheta_p^{\gamma}(C) = \frac{1}{p} \arctan \frac{\Im\gamma_p(C)}{\Re\gamma_p(C)}, \tag{9}$$

with $\Im\gamma_p$ and $\Re\gamma_p$ the imaginary and real part of $\gamma_p$, respectively. Instead of the normals $\mathbf{n}$, the descriptor is based on the position vector $\mathbf{r}$. One might be tempted to think that Equation 9 is related to the irreducible representation of $W_1^{p,0}$. This is, however, not the case as the weighting in Equation 8 is done with respect to the magnitude of $\mathbf{r}$, while in $W_1^{p,0}$ the weighting is done with respect to the contour integral. Furthermore, in contrast to the Minkowski tensors, Equation 8 takes into account only the vectors at the vertices and not the vectors along the full cell contour. While this seems to be a technical detail, it has severe consequences. The theoretical basis, which guarantees continuity, no longer holds. This leads to unstable behaviour, as illustrated in **Figure 6**. While the shape of the polygons is almost identical, there is a jump in $\gamma_3$ and $\gamma_4$ as we go from four to three vertices. The weighting according to the magnitude cannot cure this, as the magnitude of the shrinking vector is far from zero shortly before this vertex vanishes. In the context of cellular systems, such deformations are a regular occurrence rather than an artificially constructed test case. We will demonstrate the impact while discussing the results and recommend only using robust descriptors such as the Minkowski tensors or their irreducible representations.

We further note that bond order parameters $\Psi_p^{bond}$, which consider the connection between cells, have also been used as shape descriptors (Li and Ciamarra, 2018; Durand and Heu, 2019; Pasupalak et al., 2020). Introduced in Nelson and Halperin (1979), these parameters are

$$\Psi_p^{bond} = \frac{1}{B} \sum_{b=1}^{B} e^{ip\theta_b}, \tag{10}$$

with $B$ denoting the number of bonds and $\theta_b$ the orientation of the bond $b$. In the context of mono-layer tissues, $B$ is understood as the number of neighbors, and $\theta_b$ as the orientation of the connection of the center of mass of the current cell with the center of mass of the neighbor $b$ (Loewe et al., 2020; Monfared et al., 2023). So cells with the same contour but different neighbor relations lead to different bond order parameters. This characteristic should already disqualify these measures as shape descriptors. However, as discussed in detail in Mickel et al. (2013), they are not robust even for the task they are designed for. We therefore do not discuss them further. The same argumentation holds for other measures which are based on connectivity, as, e.g., considered in (Graner et al., 2008; Merkel et al., 2017).

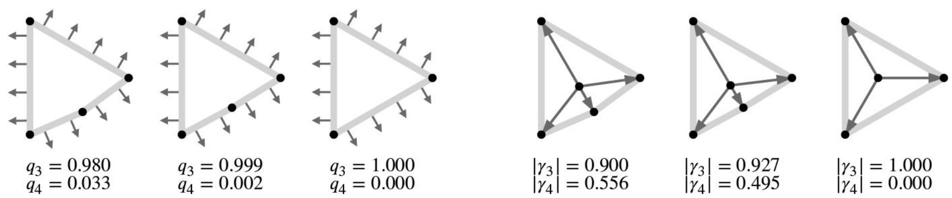



**Figure 6.**

### Defining *p*-atic order for deformable objects requires robust shape descriptors.

Shown is the strength of $p$-atic order for a polygon converging to an equilateral triangle. a) using $q_p$ and b) using $\gamma_p$. The considered vectors used in the computations, normals $\mathbf{n}$ of the contour for the Minkowski tensors and $\mathbf{r}_i$ for $\gamma_p$, are shown. Note that the removal of the forth vertex highly influences the value of $\gamma_p$. How $\mathbf{x}_c$ is calculated - as the mean of the vertex coordinates or as the center of mass of the polygon - can also slightly alter the results. We here used the described approach following Armengol-Collado et al. (2023).

### Coarse-grained quantities

We define coarse-grained quantities, following closely the strategy used in Armengol-Collado et al. (2023). We therefore regard the coarse-grained strength of $p$-atic order $Q_p = Q_p(\mathbf{x})$, which is the average of all shape functions $\frac{\psi_p}{\psi_0}$ (or equivalently $q_p e^{ip\vartheta_p}$) of cells whose center of mass $\boldsymbol{x}$ lies within a circle with radius $R$ and center $\mathbf{x}$. In a formula this is:

$$Q_p(\mathbf{x}) = \frac{\sum_{j=1}^{N} q_p(C_j) e^{ip\vartheta_p(C_j)} \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)}{\sum_{j=1}^{N} \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)} \tag{11}$$

where $N$ is the number of cells. $\mathscr{C}_i$ denotes the $i$−th cell and Θ denotes the Heaviside step function with $\Theta(x) = 1$ for $x > 0$ and $\Theta(x) = 0$ otherwise. As in Armengol-Collado et al. (2023) the position $\mathbf{x}$ is sampled over a square grid with a spacing close to the mean cell radius $R_{\mathscr{C}}$. We calculate this as $R_C = \sqrt{\frac{A_C}{\pi}}$ with $A_{\mathscr{C}}$ the cell area. Eq. (11) provides the basis for validation of continuous $p$-atic liquid crystal theories on the tissue scale, e.g. (Giomi et al., 2022a, b). For comparison we also consider the coarse-grained shape function $\Gamma_p = \Gamma_p(\mathbf{x})$, which is the average of all shape functions $\gamma_p$, which has been considered in (Armengol-Collado et al., 2023) and reads

$$\Gamma_p(\mathbf{x}) = \frac{\sum_{j=1}^{N} \gamma_p(C_j) \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)}{\sum_{j=1}^{N} \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)}. \tag{12}$$

Following Armengol-Collado et al. (2023) $\mathbf{x}_c$ is calculated as $\mathbf{x}_c = \frac{1}{V}\sum_{i=1}^{V} \mathbf{x}_i$ for $\Gamma_p$.

While $Q_p$ and $\Gamma_p$ allow to analyze clustering on the tissue scale, their averages $\overline{Q_p}$ and $\overline{\Gamma_p}$ are tightly related to the statistical properties of the probability distributions of $q_p$ and $|\gamma_p|$. We consider these properties only for comparison and follow the method used in Armengol-Collado et al. (2023): At first we calculate for every time instance/frame the spatial means, $Q_p^t$ and $\Gamma_p^t$, by averaging over all grid points. Then we calculate $\overline{Q_p}$ and $\overline{\Gamma_p}$ by averaging in time, so averaging over all $Q_p^t$ and $\Gamma_p^t$. As in Armengol-Collado et al. (2023) the s.e.m. and the standard derivation refer to the averaging in time. $\overline{Q_p}$ and $\overline{\Gamma_p}$ will be used for the discussion of a proposed hexatic-nematic crossover at larger length scales (Armengol-Collado et al., 2023).

## Simulation methods

We consider two modeling approaches, an active vertex model and a multiphase field model. Both have been proven to capture various generic properties of epithelial tissue (Hakim and Silberzan, 2017; Alert and Trepat, 2020; Moure and Gomez, 2021). The key aspects of formulations are outlined in ***Box 1*** and ***Box 2***. They refer to (Koride et al., 2018; Das et al., 2021; Killeen et al., 2022) and (Wenzel and Voigt, 2021; Jain et al., 2023; 2024b), respectively. Parameters used within these models are listed in **Appendix 1 Table 1** and **Appendix 1 Table 2**.

The initial configuration for the active vertex model simulations was built by placing $N$ points randomly in a periodic simulation box of length $L \times L$ and ensuring that no two points were within a distance less than $r_{\text{cut}}$ from each other. The points were used to create a Voronoi tiling. Typically, such a tiling had cells of very irregular shapes. To make cell shapes more uniform, the centroids of each tile were computed and used as seeds for a new Voronoi tiling. This process was iterated until it converged within the tolerance of $10^{-5}$, resulting in a so-called well-centered Voronoi tiling with cells of random shapes but similar sizes. Initial directions of polarity vectors $\hat{\mathbf{n}}_{C_i}$ were chosen at random from a uniform distribution. The initial configuration for the multiphase field model simulations considers a regular arrangement of $N$ cells of equal size in a periodic simulation box of length $L \times L$ with randomly chosen directions of self-propulsion and simulating for several time steps until the cell shapes appear sufficiently irregular. For both models we consider 100 cells and analyse time instances of the evolution.

## Experimental setup

### Cell culture

Madin-Darby canine kidney (MDCK) cells were cultured in DMEM (DMEM, low glucose, GlutaMAX™ Supplement, pyruvate) supplemented with 10 % fetal bovine serum (FBS; Gibco) and 100 U/mL penicillin/streptomycin (Gibco) at 37 °C with 5 % $CO_2$. The cell line was tested for mycoplasma.

### Monolayer preparation

Cells were seeded on glass-bottom dishes (Mattek) pretreated with 10 µg/mL fibronectin (human plasma; Gibco) in phosphate-buffered saline (PBS, pH 7.4; Gibco). Fibronectin was incubated for 30 min at 37 °C. The initial cell seeding density was sparse. The sample was imaged approximately 24 hours later, when a confluent monolayer had formed.

### Live cell imaging

The sample was imaged using a Nikon ECLIPSE Ti microscope equipped with an H201-K-FRAME chamber, heating system (Okolab), and $CO_2$ pump (Okolab), which maintained environmental conditions at 37 °C and 5 % $CO_2$. Phase-contrast images were acquired using a 10×, NA=0.3 Plan Fluor objective and an Andor Neo 5.5 sCMOS camera.

### External dataset

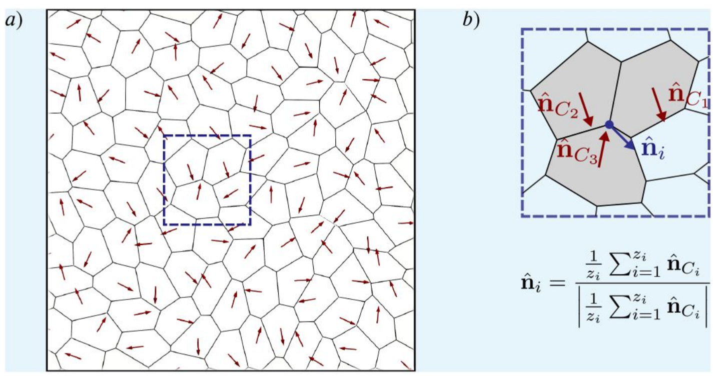
As a complementary approach, we analyzed data from the study by Armengol-Collado et al. Armengol-Collado et al. (2023), which included images of MDCK GII monolayers labeled for E-cadherin. These images were made publicly available by the authors via GitHub.

**Box 1.**

**Active vertex model**

The active vertex model is based on models discussed in (Koride et al., 2018; Das et al., 2021; Killeen et al., 2022). A confluent epithelial tissue is appreciated as a two-dimensional tiling of a plane with the elastic energy given as (Honda, 1983; Farhadifar et al., 2007; ***Honda and Nagai, 2022***)

$$E_{\mathrm{VM}} = \frac{K_A}{2} \sum_C \left(A_C - A_0\right)^2 + \frac{K_P}{2} \sum_C \left(P_C - P_0\right)^2, \tag{13}$$




**Box 1—figure 1.**

a) Cell contour of the active vertex model. Red arrows represent the polarity vectors that set each cell's instantaneous direction of self-propulsion. b) Zoom in on a vertex surrounded by three cells showing how the direction of self-propulsion on a vertex is calculated.

where $A_C$ and $P_C$ are, respectively, area and perimeter of cell $C$, $A_0$ and $P_0$ are, respectively, preferred area and perimeter, and $K_A$ and $K_P$ are, respectively, area and perimeter moduli. For simplicity, it is assumed that all cells have the same values of $A_0$, $P_0$, $K_A$, and $K_P$. The model can be made dimensionless by dividing Equation 13 by $e^* = K_A A_0^2$,

$$e_{\mathrm{VM}} = \frac{1}{2}\sum_C \left(a_C - 1\right)^2 + \frac{k_P}{2}\sum_C \left(p_C - p_0\right)^2, \tag{14}$$

where $a_C = A_C / A_0$ (i.e. $l^* = \sqrt{A_0}$ is the unit of length), $p_c = P_C/\sqrt{A_0}$, $k_P = K_P/(K_A A_0)$, and $p_0 = P_0/\sqrt{A_0}$ is called the shape index, and it has been shown to play a central role in determining if the tissue is in a solid or fluid state (Bi et al., 2015).

One can use Equation 13 to find the mechanical force on a vertex $i$ as $\mathbf{F}_i = -\nabla_{\mathbf{r}_i} E_{\mathrm{VM}}$, where $\nabla_{\mathbf{r}_i}$ is the gradient with respect to the position $\mathbf{r}_i$ of the vertex. The expression for $\mathbf{F}_i$ is given in terms of the position of the vertex $i$ and its immediate neighbours (Tong et al., 2023), which makes it fast to compute. Furthermore, cells move on a substrate by being noisily self-propelled along the direction of their planar polarity described by a unit-length vector $\hat{\mathbf{n}}_{C_i}$ from which the polarity direction at the vertex $\hat{\mathbf{n}}_i$ is defined, see **Box 1 - figure 1 b**), where $z_i$ is the number of neighbours of vertex $i$. It is convenient to write $\hat{\mathbf{n}}_i = \cos(\alpha_i)\mathbf{e}_x + \sin(\alpha_i)\mathbf{e}_y$, where $\alpha_i$ is an angle with the $x$−axis of the simulation box. The equations of motion for vertex $i$ is a force balance between active, elastic, and frictional forces and are given as

$$\dot{\mathbf{r}}_i = v_0 \hat{\mathbf{n}}_i + \frac{1}{\gamma_{fr}}\mathbf{F}_i, \quad \dot{\alpha}_i = \xi_i(t), \tag{15}$$

where $v_0$ is the magnitude of the self-propulsion velocity defined as $f_0/\gamma_{fr}$. Furthermore the overdot denotes the time derivative, $\gamma_{\mathrm{fr}}$ is the friction coefficient, $f_0$ is the magnitude of the self-propulsion force (i.e., the activity), $\xi_i(t)$ is Gaussian noise with $\langle \xi_i(t) \rangle = 0$ and $\langle \xi_i(t)\xi_j(t') \rangle = 2D_{\mathrm{r}}\delta_{ij}\, \delta(t - t')$, where $D_{\mathrm{r}}$ is the rotation diffusion constant and $\langle \cdot \rangle$ is the ensemble average over the noise. Equations of motion can be non-dimensionalised by measuring time in units of $t^* = \gamma_{\mathrm{fr}}/(K_A A_0)$ and force in units of $f^* = K_A A_0^{3/2}$ and integrated numerically using the first-order Euler-Maruyama method with timestep $\tau$.

**Box 2.**

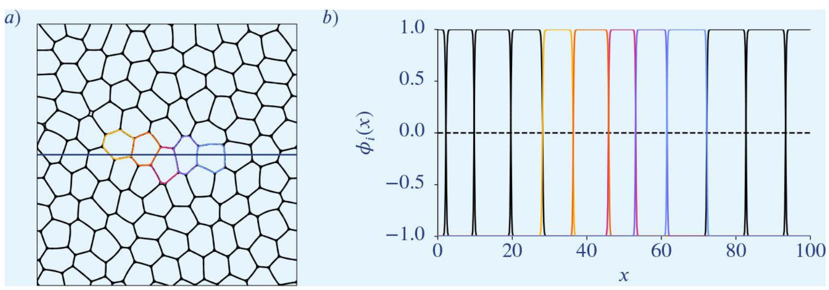
**Multiphase field model**

The multiphase field modelling approach follows Wenzel and Voigt (2021); Jain et al. (2023, 2024b). Each cell is described by a scalar phase field variable $\Phi_i$ with $i = 1, 2, \ldots, N$, where the bulk values $\Phi_i \approx 1$ denotes the cell interior, $\Phi_i \approx -1$ denotes the cell exterior, and with a diffuse interface of width $\mathcal{O}(\epsilon)$ between them representing the cell boundary. The phase field $\Phi_i$ follows the conservative dynamics

$$\partial_t \phi_i + \mathbf{v}_i \cdot \nabla \phi_i = \Delta \frac{\delta \mathcal{F}}{\delta \phi_i}, \tag{16}$$

with the free energy functional $\mathcal{F} = \mathcal{F}_{CH} + \mathcal{F}_{INT}$ containing the Cahn-Hilliard energy

$$\mathcal{F}_{CH} = \frac{1}{Ca}\sum_{i=1}^{N}\int_\Omega \left(\frac{\epsilon}{2}\|\nabla\phi_i\|^2 + \frac{1}{4\epsilon}(\phi_i^2 - 1)^2\right) d\mathbf{x}, \tag{17}$$




**Box 2—figure 1.**

a) Cell contours of the multiphase field model. b) Corresponding phase field functions along the horizontal line in a). Colours correspond to the once in a).

where the Capillary number $Ca$ is a parameter for tuning the cell deformability, and an interaction energy consisting of repulsive and attractive parts defined as

$$\mathcal{F}_{INT} = \frac{1}{In}\sum_{i=1}^{N}\int_{\Omega}\sum_{j\neq i} a_r(\phi_i+1)^2(\phi_j+1)^2 - a_a(\phi_i^2-1)^2(\phi_j^2-1)^2 d\mathbf{x}, \tag{18}$$

where $In$ denotes the interaction strength, and $a_r$ and $a_a$ are parameters to tune contribution of repulsion and attraction. The repulsive term penalises overlap of cell interiors, while the attractive part promotes overlap of cell interfaces. The relation to former formulations is discussed in Happel and Voigt (2024).

Each cell is self-propelled, and the cell activity is introduced through an advection term. The cell velocity field is defined as

$$\mathbf{v}_i(\mathbf{x},t) = v_0\mathbf{e}_i(t)\hat{\phi}_i(\mathbf{x},t), \tag{19}$$

where $v_0$ is used to tune the magnitude of activity, $\hat{\phi}_i = \frac{\phi_i+1}{2}$ and $\mathbf{e}_i = [\cos\theta_i(t), \sin\theta_i(t)]$ is its direction. The migration orientation $\theta_i(t)$ evolves diffusively with a drift that aligns to the principal axis of cell's elongation as $d\theta_i = \sqrt{2D_r}dW_i(t) + \alpha(\beta_i(t)-\theta_i(t))dt$, where $D_r$ is the rotational diffusivity, $W_i$ is the Wiener process, $\beta_i(t)$ is the orientation of the cell elongation and $\alpha$ controls the time scale of this alignment.

The resulting system of partial differential equations is considered on a square domain with periodic boundary conditions and is solved by the finite element method within the toolbox AMDiS (Vey and Voigt, 2007; Witkowski et al., 2015) and the parallelization concept introduced in Praetorius and Voigt (2018) is considered, which allows scaling with the number of cells. We in addition introduce a de Gennes factor in $\mathcal{F}_{CH}$ to ensure $\Phi_i \in [-1, 1]$, see Salvalaglio et al. (2021).

## Image analysis

All image data, including both our phase-contrast recordings and the E-cadherin-labeled external dataset, were analyzed using Cellpose Stringer et al. (2021). Segmentation was performed using manually trained models, each tailored to its respective dataset.

## Extraction of the contour

Starting out from the segmented images (stored as grayscale images) we extract the contour of the cells using the python package *scikit-image* (van der Walt et al., 2014). To get rid of pixel-shaped artifacts we smooth the contour by replacing every coordinate by the average over 9 neighboring points in the outline.

# Results

Quantifying orientational order in biological tissues can be realized by Minkowsky tensors. The orientation $\vartheta_p$ and the strength of $p$-atic order $q_p$ in Equation 7 can be computed for each cell. Minkowski tensors provide reliable quantities describing how cell shapes align with specific rotational symmetries. As already indicated in **Figure 2** situations might occur in which rotational symmetries can not be associated with one specific $p$, but various symmetries seem to be present at the same time. One might be tempted to compare the probability distribution functions (PDFs) of $q_p$ or the mean values $\overline{q_p}$ for different $p$ in order to identify a dominating $p$-atic order. This is particularly important for interpreting nematic ($p$ = 2) and hexatic ($p$ = 6) orders, which describe distinct symmetries but have been found to coexist in biological systems. However, a direct comparison of these quantities only makes sense if they are comparable to each other. As already mentioned in **Figure 4** this is not the case, as, e.g., $q_2 = 1.0$ cannot be realized for a cell with a given area. Another question one might ask is if these values are independent of each other. To answer this question, we first address statistically if $q_2$ and $q_6$, evaluated for each cell, are independent. Second, we explore how $q_p$ depends on key parameters determining tissue mechanics. In a third step we coarsegrain these quantities using the measures $Q_2$ and $Q_6$ in **Equation 11**. With these quantities we address a proposed hexatic-nematic crossover in epithelial tissue, where hexatic order dominates at small scales and nematic order prevails at larger scales (Eckert et al., 2023; Armengol-Collado et al., 2023, 2024). For these three tasks we analyze simulation data from two computational models, the active vertex model ***Box 1*** and the multiphase field model ***Box 2***. Although these models differ conceptually, both have been validated in studies of cell monolayer mechanics, including solid-liquid transitions, neighbor exchange dynamics, and stress profiles (Fletcher et al., 2014; Alt et al., 2017; Li et al., 2019; Balasubramaniam et al., 2021; Wenzel and Voigt, 2021; Sknepnek et al., 2023; Melo et al., 2023). Key model parameters are the deformability and the activity strength. They are crucial for capturing the coarse-grained properties of confluent tissues (Jain et al., 2023, ***2024b***). In the active vertex model, deformability is controlled by the target shape index $p_0$, reflecting the balance between cell-cell adhesion and cortical tension. The multiphase field model, by contrast, encodes deformability through the capillary number $Ca$, which directly incorporates cortical tension. We vary activity strength ($v_0$), the shape index ($p_0$), and the capillary number ($Ca$), ensuring all parameter combinations remain in the fluid regime. Fluidity was confirmed using by mean square displacement (MSD) (Loewe et al., 2020), neighbor number variance (Wenzel and Voigt, 2021), and the self-intermediate scattering function (Bi et al., 2016).

These studies demonstrate independence of $q_2$ and $q_6$, a general trend of increasing $\overline{q_2}$ and decreasing $\overline{q_6}$ for higher activity and deformability, and a consistent decrease of $\overline{Q_2}$ and $\overline{Q_6}$ for increasing coarse-graining radius $R$. However, as $q_2$ and $q_6$ are not directly comparable and $q_2$ and $q_6$ (and therefore also $Q_2$ and $Q_6$) are independent, the concept of a hexatic-nematic transition - typically requiring a single order parameter - may not be applicable in this context. Even if the proposed hexatic-nematic crossover (Armengol-Collado et al., 2023) is not a formal phase transition, one might expect it to be quantifiable. Yet, its characterization appears to depend strongly on the maximum attainable value of $q_2$, which in turn is influenced by several parameters. To further explore this, we reanalyze the experimental data for MDCK cells from (Armengol-Collado et al., 2023) using Minkowski tensors and compute $q_2$ and $q_6$ as defined in Equation 7. Our analysis also indicates independence of $q_2$ and $q_6$ supporting the same interpretation as above. The statistical properties and probability distributions of these data remain consistent when full cellular boundaries from microscopy images are used. However, an increase in hexatic order ($p = 6$) is observed when cell shapes are approximated by polygons. This suggests that the dominant hexatic order reported at the cellular scale in Armengol-Collado et al. (2023) may stem from the geometric simplification of cell boundaries. We also compute $Q_2$ and $Q_6$ (Equation 11) and again observed a consistent decrease with increasing coarse-graining radius $R$, without evidence of a measurable hexatic-nematic crossover. To better understand the differences with the findings in Armengol-Collado et al. (2023) we further analyze both simulation and experimental data using the alternative shape measures $\gamma_p$ (Equation 8) and $\Gamma_p$ (Equation 12) considered in Armengol-Collado et al. (2023). These measures reproduce the reported results.

## Independence of $q_2$ and $q_6$

We examine the distribution of ($q_2$, $q_6$) values across deformability-activity parameter pairs in both computational models (**Figure 7**). Both models show consistent trends: In near-solid regimes (**Figure 7** lower left), $q_2$ and $q_6$ values cluster tightly due to restricted shape fluctuations. However, even in this regime, small $q_2$ values can correspond to either small or large

$q_6$ values, and vice versa. In more fluid-like regimes, with higher activity and higher deformability (**Figure 7** upper right), $q_2$ and $q_6$ values become highly scattered. Each $q_2$ value spans a broad range of $q_6$ values, and vice versa, indicating their independence. In order to quantify this we compute the distance correlation (Székely et al., 2007), which is a statistical measure quantifying linear and non-linear dependency in given data. Thereby a value of 0.0 corresponds to independence whereas a value of 1.0 corresponds to a strong dependence between the datasets. As can be seen in **Figure 7—figure Supplement 1** the obtained distance correlation for $q_2$ and $q_6$ is quite low, underscoring that these quantities are independent. Furthermore the corresponding P-values, as shown in **Figure 7—figure Supplement 2** are mostly larger than 0.1, indicating that the weak correlation found in **Figure 7— figure Supplement 1** is not significant. This leads to the conclusion that $q_2$ and $q_6$ measure distinct aspects of cell shape anisotropy. As a consequence both orders, nematic and hexatic, need to be considered independently. There cannot be a single parameter which describes a crossover between both.

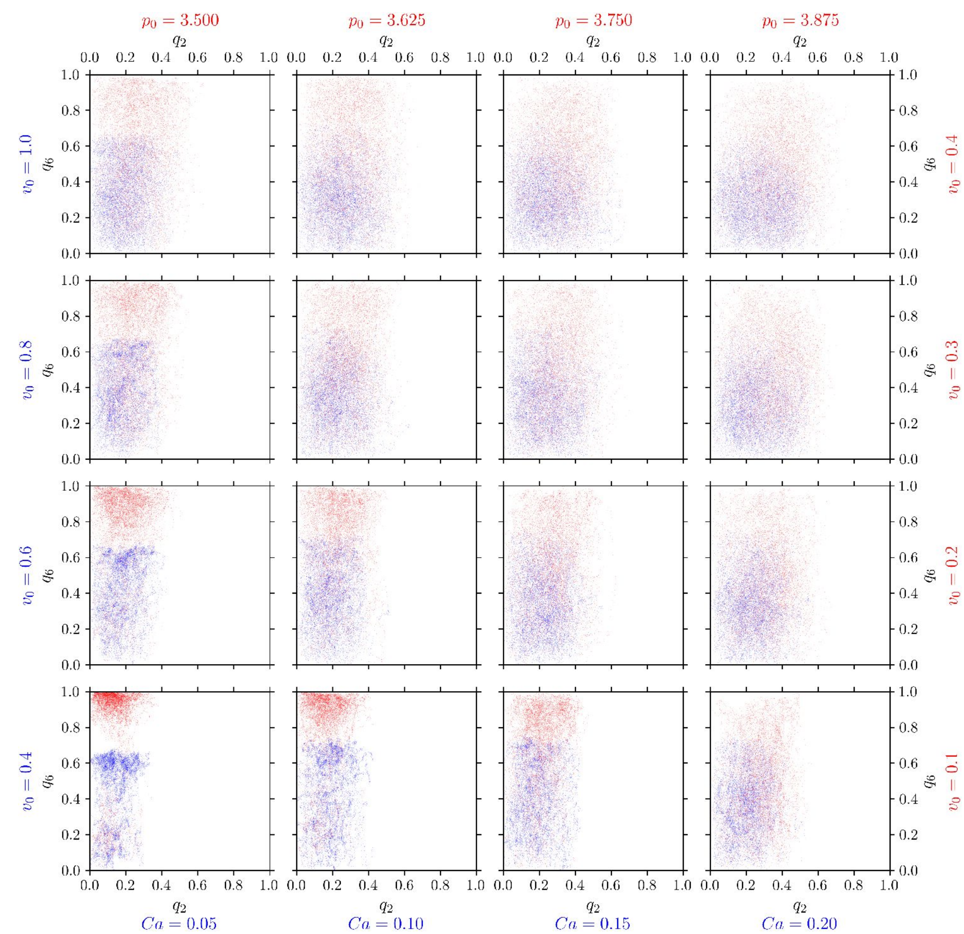
A full investigation of potential dependencies between $q_p$ for arbitrary combinations of $p$'s resulting, e.g. from symmetry arguments is beyond the scope of this paper.

## Dependence of $q_p$ on activity and deformability

We now explore how tissue properties such as activity and deformability influence cell shape and orientational order. We again focus on nematic ($p$ = 2) and hexatic ($p$ = 6) orders, as shown in **Figure 8—figure Supplement 1** and **Figure 8—figure Supplement 2**. Additional results for $p$ = 3, 4, 5 are provided in **Appendix 2-Figure 14**– **Appendix 2-Figure 16** (fixed activity) and **Appendix 2-Figure 17**– **Appendix 2-Figure 19** (fixed deformability). In all plots all cells and all time steps are considered. In **Figure 8—figure Supplement 1**, we vary deformability ($p_0$ in the active vertex model and $Ca$ in the multiphase field model) while keeping the activity $v_0$ constant. Results are presented for the active vertex model (left column) and the multiphase field model (right column). Activity increases from bottom to top rows. Both models show qualitatively similar trends in the probability distribution functions (PDFs) of $q_2$ and $q_6$. For $p$ = 6, increasing deformability shifts the PDF of $q_6$ to the left, indicating lower mean values. In contrast, for $p$ = 2, higher deformability leads to higher $q_2$ values, reflected in a rightward shift of the PDF. This trend is confirmed by the mean values, shown as a function of deformability ($p_0$ and $Ca$, respectively) in the inlets. Additionally, the PDFs broaden with increasing deformability, and this effect is more pronounced at lower activity levels. A notable difference between the models is the range of $q_p$ values. In the multiphase field model, $q_p$ rarely exceeds 0.8 due to the smoother, more rounded cell shapes, whereas the active vertex model often produces higher $q_p$ values. In **Figure 8—figure Supplement 2**, deformability is held constant while activity $v_0$ is varied. Results are again shown for the active vertex model (left column) and the multiphase field model (right column), with increasing activity indicated by brighter colors. Deformability increases from bottom to top rows. The trends mirror those observed in **Figure 8— figure Supplement 1**. Increasing activity reduces the mean value of $q_6$ while increasing $q_2$, see inlets. The effects of activity are more pronounced at lower deformabilities; at higher deformabilities, differences between parameter regimes diminish. The overall behavior of $q_2$ and $q_6$ is summarized in **Figure 8**, which shows the mean values $\overline{q_2}$ and $\overline{q_6}$ as functions of activity and deformability. For both models:

- $\overline{q_2}$ increases with higher activity or deformability
- $\overline{q_6}$ decreases with higher activity or deformability.

This behavior is consistent with a trend towards nematic order in more dynamic regimes and towards hexatic order in more constrained, less dynamic regimes. The first is further confirmed by recent findings in analysing T1 transitions and their effect on cell shapes Jain et al., 2024a). These studies suggest that cells transiently elongate when they are undergoing T1 transitions. As




## Figure 7.

### Nematic ($p$ = 2) and hexatic ($p$ = 6) order are independent of each other.

$q_6$ (y-axis) versus $q_2$ (x-axis) for all cells in the multiphase field model (blue) and active vertex model (red). For each cell and each timestep we plot one point ($q_2$, $q_6$). Each panel corresponds to specific model parameters; $Ca$ and $v_0$ for multiphase field model, and $p_0$ and $v_0$ for the active vertex model, representing deformability and activity, respectively.

**Figure 7—figure supplement 1**. Distance correlation $dCor(q_2, q6)$ between $q_2$ and $q_6$

**Figure 7—figure supplement 2**. P-values of the distance correlation $P_{dCor(q_2,q6)}$ between $q_2$ and $q_6$

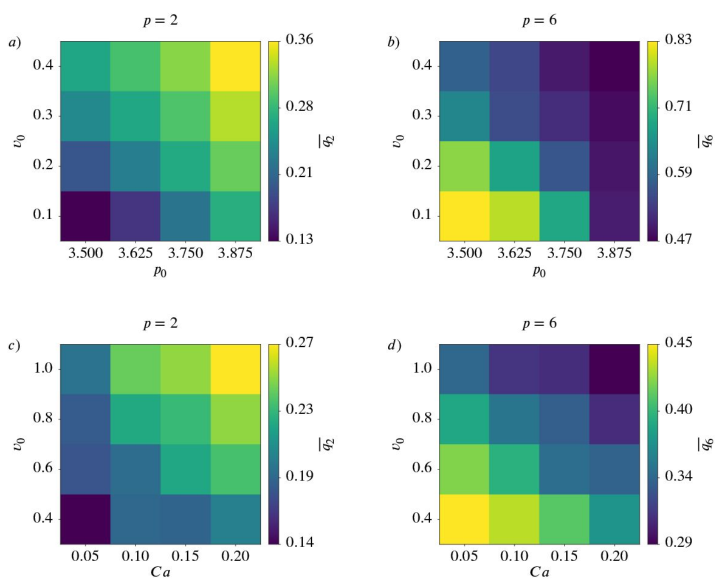



**Figure 8.**

**Nematic ($p$ = 2) and hexatic ($p$ = 6) order depend on activity and deformability of the cells.**

Mean value $\overline{q_p}$ for $p$ = 2 (left) and $p$ = 6 (right) as function of deformability $p_0$ or $Ca$ and activity $v_0$ for active vertex model (a) and b)) and multiphase field model (c) and d)).

**Figure 8—figure supplement 1**. Nematic ($p$ = 2) and hexatic ($p$ = 6) order depend on deformability of the cells

**Figure 8—figure supplement 2**. Nematic ($p$ = 2) and hexatic ($p$ = 6) order depend on activity of the cells

the number of T1 transitions increases with activity or deformability Jain et al., 2024b) this elongation contributes to the observed behavior. The second is consistent with the emergence of hexagonal arrangements in solid-like states.

Corresponding results for $p$ = 3, 4, 5 are shown in **Appendix 2-Figure 20**. While $q_3$, $q_4$, $q_5$ also increase with increasing activity or deformability, the dependency is not as pronounced as for $q_2$.

## Coarse-grained quantities $Q_2$ and $Q_6$ and potential hexatic-nematic crossover

For every parameter configuration in the active vertex model (**Figure 9—figure Supplement 1**) and in the multiphase field model (**Figure 9—figure Supplement 2**) we compute the coarse-grained quantities $Q_2$ and $Q_6$ for various coarse-graining radii $R$. One might ask the question if the observed trends for $\overline{q_2}$ and $\overline{q_6}$ for higher activity and deformability in **Figure 8** are also present on larger scales. We therefore investigate the behavior of $\overline{Q_2}$ and $\overline{Q_6}$ upon varying activity and deformability, see **Figure 9**. This is exemplified for $R/R_{cell}$ = 8.0. The trends seen for $\overline{q_2}$ and $\overline{q_6}$ - so increasing ( $\overline{q_2}$ ) or decreasing ( $\overline{q_6}$ ) with higher activity or deformability - are lost for $\overline{Q_2}$ and less pronounced for $\overline{Q_6}$ .

In Armengol-Collado et al. (2023) a similar approach was used to identify a hexatic-nematic crossover, see (Armengol-Collado et al., 2023, **Fig. 3e**). This cross-over is considered at the coarse-graining radius at which the two curves for $\overline{Q_2}$ and $\overline{Q_6}$ as a function of $R$ cross. While already conceptually questioned above we consider these investigations to compare with Armengol-Collado et al. (2023). However, regardless of the model there is no consistent trend indicative of a potential crossover. These results question the proposed hexatic-nematic crossover reported in (Armengol-Collado et al., 2023). To further explore this issue we next test the existence of such a crossover directly on the data considered in (Armengol-Collado et al., 2023).

## Analyzing experimental data for MDCK cells

The experimental data for confluent monolayers of MDCK GII cells used in (Armengol-Collado et al., 2023) are provided in two different formats, as microscopy images and as polygon data with the calculated vertex points per cell. We consider both formats and all 68 provided configurations. The polygonal data are directly used to compute $q_2$ and $q_6$. The experimental microscopy images are segmented and the extracted cell boundaries are used to compute $q_2$ and $q_6$.

We compute the probability distribution functions (PDFs) of $q_2$ and $q_6$, **Figure 10** a) for both data sets. While they are similar if the full cellular contour from the microscopy images is considered the PDFs strongly differ if the polygonal shapes are used. The dominating hexatic ($p$ = 6) order is therefore just a consequence of the approximation of the cell boundaries by polygons. This different behavior, which shows larger values for $q_6$ for the polygonal shapes has the same origin as the difference between the regular and rounded shapes in **Figure 4**. Further differences result from the different accessible parameter range. While $q_6$ = 1.0 is possible for a perfect hexagon, $q_2$ = 1.0 cannot be realised, as this would correspond to a line.

We also test the values for $q_2$ and $q_6$ for independence, see **Figure 10** b) and in the corresponding statistical measures **Figure 10—figure Supplement 2** and **Figure 10—figure Supplement 3**. The indicated independence in the scatter plots **Figure 10** b) is confirmed by the distance correlation and the P-values, as in **Figure 7**, **Figure 7—figure Supplement 1** and **Figure 7—figure Supplement 2**. This holds for the polygonal shapes as well as for the more detailed shapes from the microscopy images.

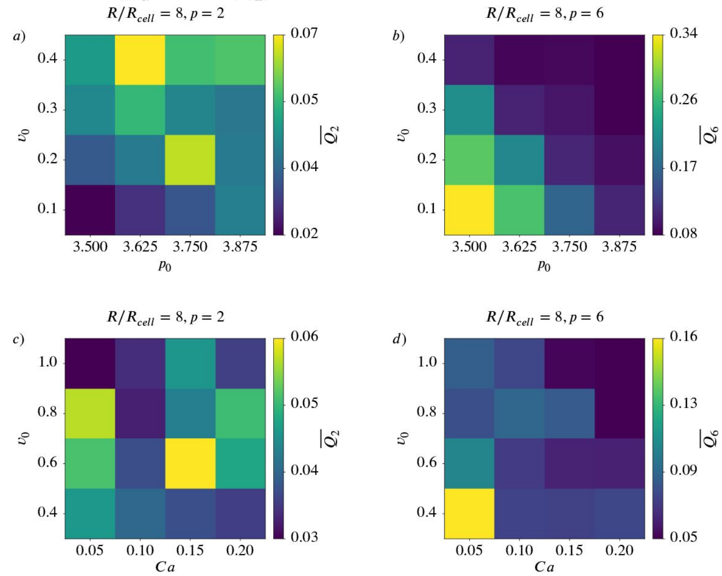
## Figure 9.

### Coarse-gained nematic ($p$ = 2) and hexatic ($p$ = 6) order for $R/R_{cell}$ = 8 depend on activity and deformability of the cells.

Mean value $\overline{Q_p}$ for $p$ = 2 (left) and $p$ = 6 (rigth) as function of deformability $p_0$ or $Ca$ and activity $v_0$ for active vertex model ($a$) and $b$)) and multiphase field model ($c$) and $d$)).

**Figure 9—figure supplement 1**. $\overline{Q_6}$ versus $\overline{Q_2}$ for different coarse graining radii in the active vertex model.

**Figure 9—figure supplement 2**. $\overline{Q_6}$ versus $\overline{Q_2}$ for different coarse graining radii in the multiphase field model.




## Figure 10.

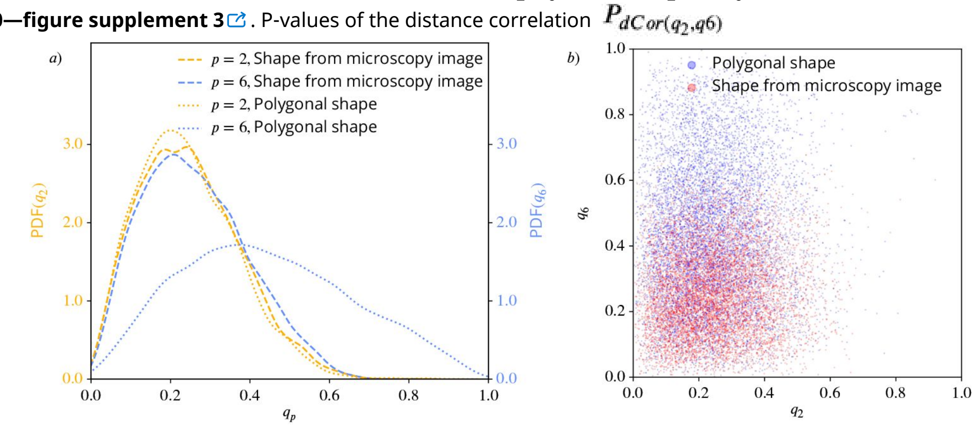
### Nematic ($p$ = 2) and hexatic ($p$ = 6) order for the cells in the experiments from Armengol-Collado et al. (2023).

$a$): Probability distribution functions (PDFs) using kde-plots, for $q_2$ (yellow) and $q_6$ (blue), once using the polygonal approximation of the cell shape and once using the detailed cell outline obtained from the microscopy pictures. $b$): $q_6$ (y-axis) versus $q_2$ (x-axis) for all cells from the experimental data in Armengol-Collado et al. (2023), once using the polygonal approximation of the cell shape (blue) and once using the detailed cell outline obtained from the microscopy pictures (red). For each cell and each timestep we plot one point ($q_2$, $q_6$).

**Figure 10—figure supplement 1**. Experimental image, segmented cell outline and polygonal shape

**Figure 10—figure supplement 2**. Distance correlation $dCor(q_2, q_6)$ between $q_2$ and $q_6$

**Figure 10—figure supplement 3**. P-values of the distance correlation $P_{dCor(q_2,q6)}$



As a consequence, the same arguments as discussed above also hold for the experimental data and thus caution against interpretation of $q_2$ and $q_6$ or their coarse-grained quantities $Q_2$ and $Q_6$ as interdependent order parameters. However, in order to compare with Armengol-Collado et al. (2023) we next compute the averaged coarse-grained quantities $\overline{Q_2}$ and $\overline{Q_6}$ for various coarse-graining radii $R$. In **Figure 11** these curves are shown for the polygonal shapes (a)) and the microscopy images (b)). As in Armengol-Collado et al. (2023) we carried out the coarse-graining until the coarse-graining radius corresponds to half of the domain width. In both plots the curves for $\overline{Q_2}$ are almost identical, reflecting the similar PDFs in **Figure 10** a). For $\overline{Q_6}$ the slope is similar but the curves are shifted. A potential crossover therefore also depends on the approximation of the cells. In any case, for the considered data no consistent hexatic-nematic crossover can be observed.

In order to resolve the discrepancy of these results with Armengol-Collado et al. (2023) we next examine the analysis using the alternative shape measures $\gamma_p$ in Equation 8, which have been considered in Armengol-Collado et al. (2023) but are shown to be not stable.

## Sensitivity of the results on the considered shape descriptor

We now demonstrate that the alternative shape descriptors $\gamma_p$ in Equation 8, which have been used in (Armengol-Collado et al., 2023), can lead to qualitatively different and thus misleading results. The corresponding figures to **Figure 1**, **Figure 2** and **Figure 4** are shown in **Appendix 3 Figure 22**, **Appendix 3 Figure 23** and **Appendix 3 Figure 21**, respectively. For the experimental data in **Figure 1** and **Figure 2** we use the Voronoi interface method (Saye and Sethian, 2011***, 2012***) to calculate the vertices of a polygon approximating the cell shape. The comparison of these figures already indicates differences between the two methods. Such differences can be seen in the approximated cell shapes, the probability distribution functions (PDFs) and more quantitatively also by comparing $\overline{|\gamma_p|}$ in **Appendix 3 Figure 23** *a*) with $\overline{q_p}$ in **Figure 2** *a*), which, e.g., for $p = 6$ almost double.

For a more detailed comparison of $|\gamma_p|$ and $q_p$ we investigate the data from the active vertex model and the polygonal approximation of the cells in Armengol-Collado et al. (2023). We restrict ourselves to this data as for multiphase field data the usage of $\gamma_p$ first requires the approximation of a cell by a polygon and we have already seen that this approximation strongly influences the results.

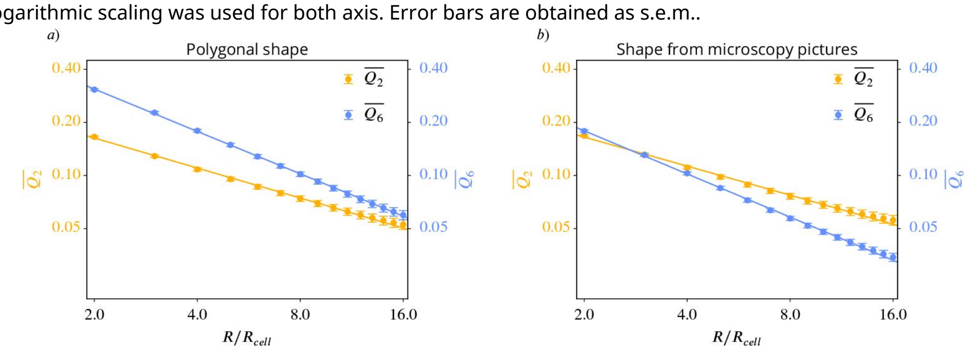
We focus on **Figure 8** and the corresponding results in **Figure 12**. Instead of the monotonic trend for activity and deformability, which was found for $\overline{q_6}$, $\overline{|\gamma_6|}$ exhibits non-monotonic trends, peaking at intermediate values of activity or deformability. For completeness we also provide the corresponding figures to **Figure 7** and **Figure 9** in **Appendix 3 Figure 24** and **Appendix 3 Figure 25**, respectively.

For the polygonal data of the MDCK cells considered in (Armengol-Collado et al., 2023), we compare the coarse-grained quantities $\overline{Q_2}$ and $\overline{Q_6}$, already considered in **Figure 11** a), with $\overline{|\Gamma_2|}$ and $\overline{|\Gamma_6|}$ computed from $\gamma_2$ and $\gamma_6$ using **Equation 12**, see **Figure 13**. For completeness we also provide the corresponding figure to **Figure 10** in **Appendix 3 Figure 26**. Besides minor differences regarding the calculation of $R_{cell}$ **Figure 13** corresponds to (Armengol-Collado et al., 2023, **Fig.3e**). Comparing **Figure 11** a) and **Figure 13** one might come to the conclusion that there is no hexatic-nematic crossover using $\overline{Q_2}$ and $\overline{Q_6}$, but there is a hexatic-nematic crossover using $\overline{|\Gamma_2|}$ and $\overline{|\Gamma_6|}$. As the only difference between these two evaluations is the considered shape characterization this analysis adds another argument that the proposed crossover is not a robust physical feature of the system.

## Figure 11.

### $\overline{Q_6}$ versus $\overline{Q_2}$ for different coarse graining radii for the experimental data from Armengol-Collado et al. (2023).

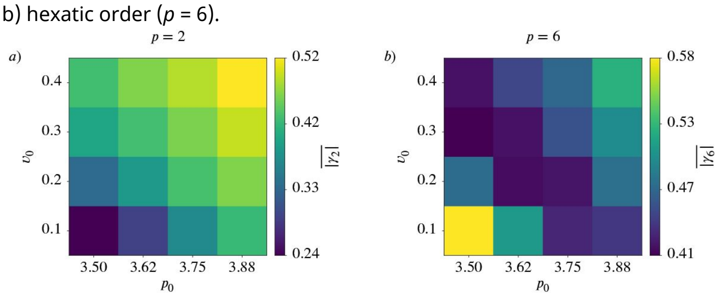
On the left side *a*) we use the polygonal approximation of the cell shape, on the right side *b*) we use the detailed cell outline obtained from the microscopy pictures. $Q_p$ was calculated according to Equation 11, the averaging of this and the choice of $R_{\mathscr{C}}$ follow the description in Coarse-grained quantities. The maximal coarse-graining radius corresponds to half the domain width. A logarithmic scaling was used for both axis. Error bars are obtained as s.e.m..




## Figure 12.

### Mean value $\overline{|\gamma_p|}$ as function of deformability $p_0$ and activity $v_0$ for active vertex model.

a) nematic order ($p$ = 2), b) hexatic order ($p$ = 6).



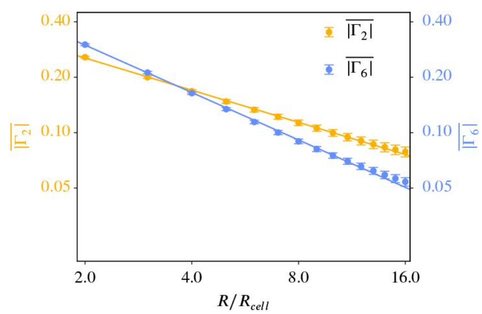



**Figure 13.**

**$\overline{|\Gamma_6|}$ versus $\overline{|\Gamma_2|}$ for different coarse graining radii for the experimental data from Armengol-Collado et al. (2023).**

We use only the polygonal approximation of the cell shape as $\gamma_p$ can only work with polygons. $Q_p$ was calculated according to Equation 12, the averaging of this and the choice of $R_{\mathscr{C}}$ follow the description in Coarse-grained quantities. The maximal coarse-graining radius corresponds to half the domain width. A logarithmic scaling was used for both axis. Error bars are obtained as s.e.m..

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These results also demonstrate that not only the approximation of the cell boundaries by polygonal shapes heavily influences the characterization of $p$-atic order but also the considered method to classify the shape might lead to qualitative different results. This confirms the argumentation in Methods and materials that Minkowski tensors should be preferred because of their stability properties.

## Discussion

In this study, we introduced Minkowski tensors as a robust and versatile tool for quantifying $p$-atic order in multicellular systems, particularly in scenarios involving rounded or irregular cell shapes. By applying this framework to extensive datasets from two distinct computational models —the active vertex model and the multiphase field model—we identified universal trends: increasing activity and deformability of the cells enhance nematic order ($p = 2$) while diminishing hexatic order ($p = 6$). The consistency of these findings across two models, despite their inherent differences, underscores the generality of our results.

While various shape characterization methods, such as the bond order parameter (Loewe et al., 2020; Monfared et al., 2023) and the shape function $\gamma_p$ (Armengol-Collado et al., 2023), have been explored in the literature, we demonstrated that the choice of shape descriptor significantly impacts the conclusions drawn. Such divergences, together with limited mathematical foundations, highlight the limitations of these alternative shape measures in capturing consistent patterns and emphasize the need for stable, reliable shape measures like the Minkowski tensors. As the stability of Minkowski tensors - in contrast to the bond order parameter or the shape function $\gamma_p$ - can be mathematically justified Minkowski tensors should be the preferred shape descriptor. This finding is not merely a technical nuance, it leads to qualitative differences. Analyzing experimental data for MDCK cells, e.g., has demonstrated that a strong hexatic order on the cellular scale has no physical origin but is a consequence of the approximation of the cell boundaries by polygonal shapes, which is a requirement to use the shape function $\gamma_6$. Considering the full cellular boundaries and $q_6$, which is derived from the Minkowski tensors, leads to a different picture, with weaker hexatic order. A critical question in the literature has been whether shape measures for different $p$-atic orders can be directly compared. While some studies have suggested relationships between $\gamma_2$ and $\gamma_6$ (Armengol-Collado et al., 2023), our results refute this notion. We demonstrated that measures like $q_2$ and $q_6$ are independent and capture fundamentally distinct aspects of cell shape and alignment. Comparing them directly is mathematically but also physically and biologically mis-leading. We further tested the hypothesis of a hexatic-nematic crossover at larger length scales by coarse-graining $q_2$ and $q_6$. To discuss such a crossover requires direct comparison of $q_2$ and $q_6$ or their coarse-grain quantities $Q_2$ and $Q_6$, which is conceptually questionable. However, the results showed no consistent trends indicative of a crossover, regardless of the considered model or the experimental data. This leads to the conclusion that the proposed hexatic-nematic crossover in (Armengol-Collado et al., 2023) is not a physical phenomena but specific to the considered method.

Our findings suggest that $p$-atic orders should be studied independently, also across length scales, as they describe complementary aspects of cellular organization. The coexistence of distinct orientational orders emphasized in different studies—such as nematic ($p = 2$) (Duclos et al., 2017; Saw et al., 2017; Kawaguchi et al., 2017), tetratic ($p = 4$) (Cislo et al., 2023), and hexatic ($p = 6$) (Li and Ciamarra, 2018)—is not contradictory but highlights the rich, multifaceted nature of cellular organization. Rather than searching for a single dominant order, future research should focus on the interplay of different $p$-atic orders and their associated defects. This suggests to not only consider $p$-atic liquid crystal theories (Giomi et al., 2022a) for one specific $p$, but combinations of these models for various $p$'s. Understanding how these orders interact may reveal how they collectively regulate morphogenetic processes.

Connecting $p$-atic orders to biological function remains a critical avenue for exploration. While the mathematical independence of $q_2$, $q_6$, and other shape measures precludes the identification of a universal dominant order, biological systems may exhibit context-dependent preferences. For example, a specific $p$-atic order might correlate with or drive a key morphogenetic event. Investigating these connections could yield insights into how tissues achieve functional organization and adapt to environmental cues. As such, while it is difficult to speak of dominating orders from a mathematical point of view, there could be a dominating order from a biological point of view, meaning the $p$-atic order connected to the governing biological process.

# Data availability

A code illustrating the extraction of the contour and the calculation from $q_p$ and $\vartheta_p$ for grayscale images can be found on Zenodo at *https://doi.org/10.5281/zenodo.15430268*.


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# Appendix 1

## Parameters for the computational models

**Appendix 1—table 1.**

**Values of the dimensionless parameters used in the active vertex model**

| Parameter | Description | Numerical Value |
|---|---|---|
| $N$ | Number of cells | 100 |
| $L$ | Simulation box size | 10 |
| $T$ | Total simulation time | 300 |
| $k_P$ | Perimeter elastic modulus | 1.0 |
| $\tau$ | Simulation time step | 0.01 |
| $v_0$ | Self-propulsion strength (i.e. activity) | 0.1 - 0.4 |
| $D_r$ | Rotation diffusion coefficient | 0.05 |
| $p_0$ | Shape index of active cells | 3.5 - 3.875 |

**Appendix 1—table 2.**

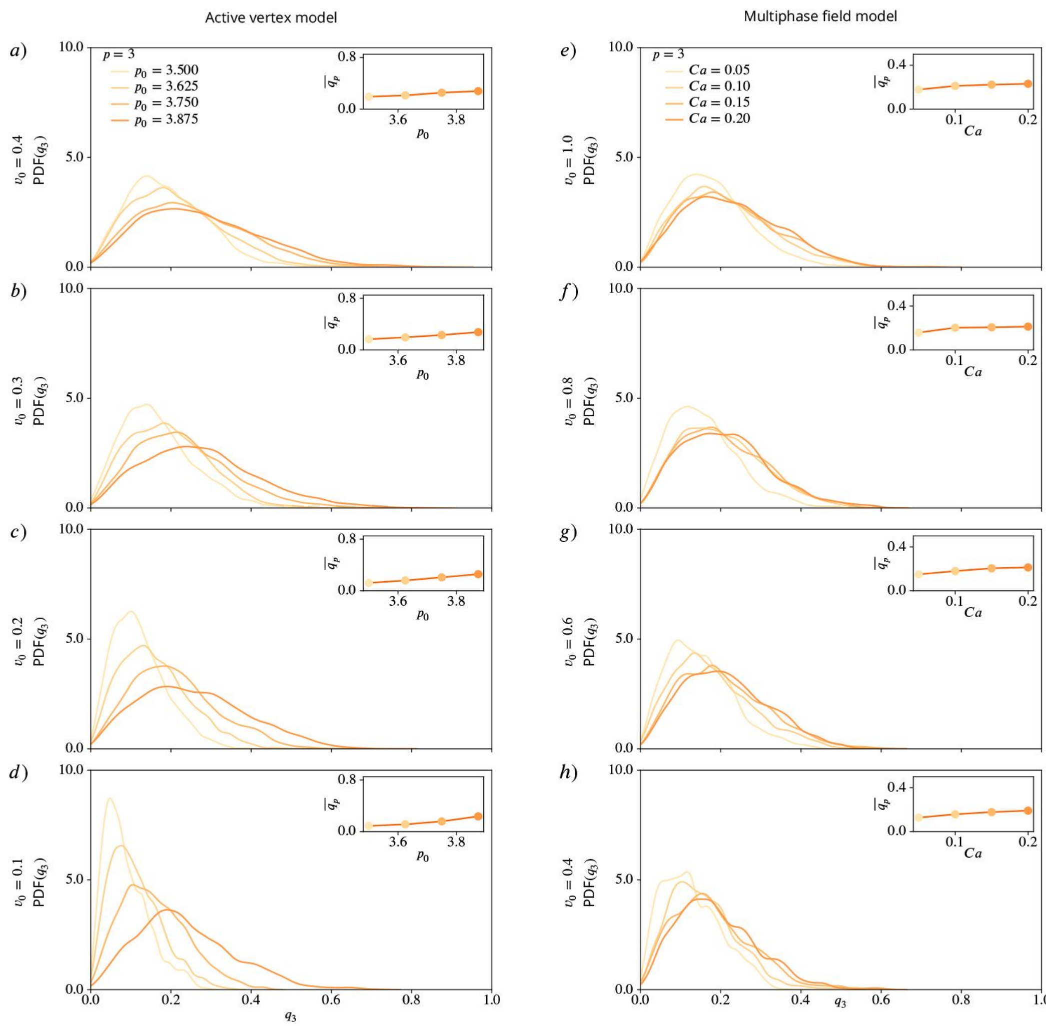
**Values of the dimensionless parameters used in the multiphase-field model simulations**

| Parameter | Description | Numerical Value |
|---|---|---|
| $N$ | Number of cells | 100 |
| $L$ | Simulation box size | 100 |
| $T$ | Total simulation time | 150 |
| $\epsilon$ | Interface width | 0.15 |
| $\tau$ | Simulation time step | 0.005 |
| $v_0$ | Self-propulsion strength (i.e. activity) | 0.4 - 1.0 |
| $D_r$ | Rotation diffusion coefficient | 0.01 |
| $\alpha$ | Alignment parameter | 0.1 |
| $Ca$ | Capillary number | 0.05 - 0.2 |
| $In$ | Interaction number | 0.1 |
| $a_a$ | Cell-cell attraction strength | 1.0 |
| $a_r$ | Cell-cell repulsion strength | 1.0 |
| $h$ | Mesh size | $5h \approx \epsilon$ |

# Appendix 2

## Results for $q_3$,$q_4$ and $q_5$

### Distribution functions in dependence of deformability



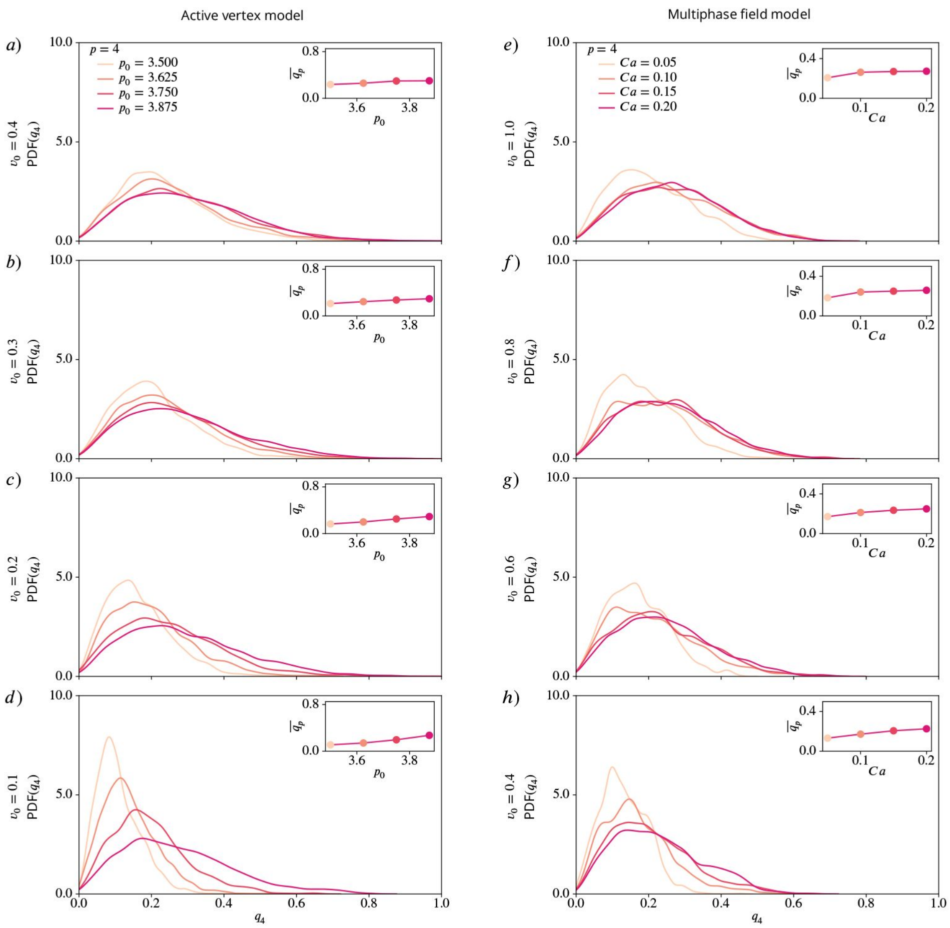

**Appendix 2—figure 14.**

**PDFs for $q_3$ using kde-plots, for varying deformability $p_0$ or $Ca$ and fixed activity $v_0$.**

Inlets show mean values of $q_3$ as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.

### Distribution functions in dependence of activity

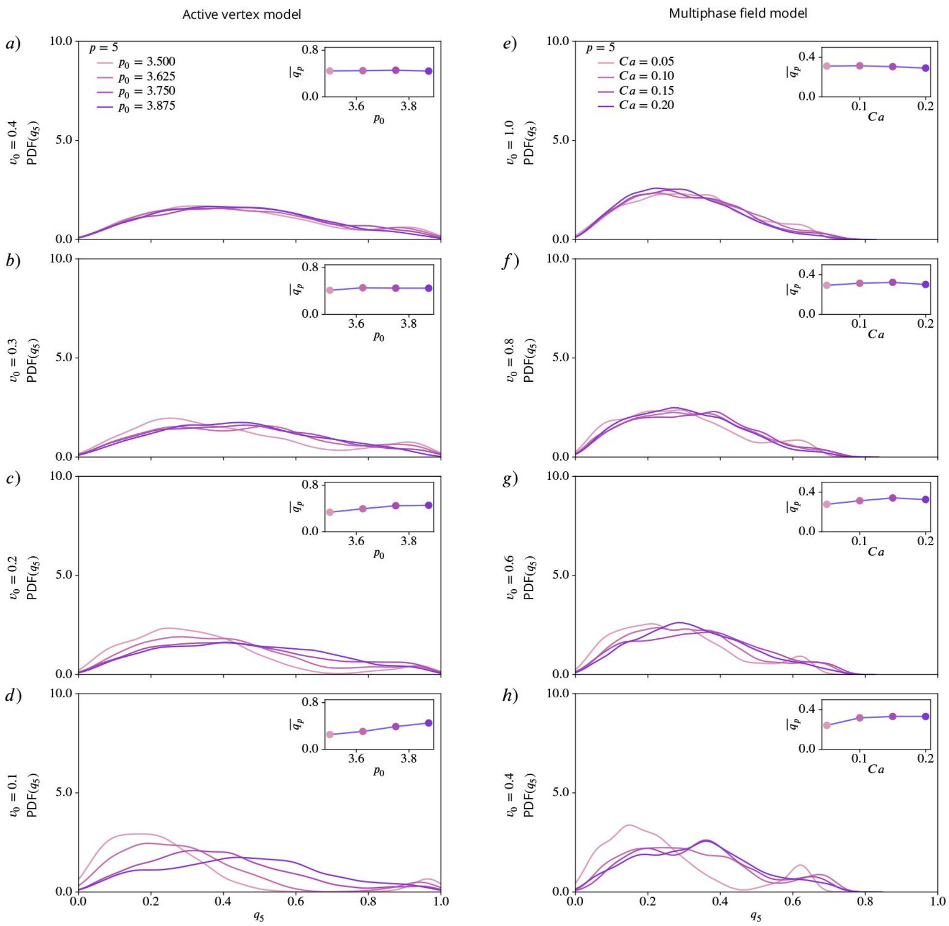
### Summarized behavior in dependence of activity and deformability




**Appendix 2—figure 15.**

**PDFs for $q_4$ using kde-plots, for varying deformability $p_0$ or $Ca$ and fixed activity $v_0$.**

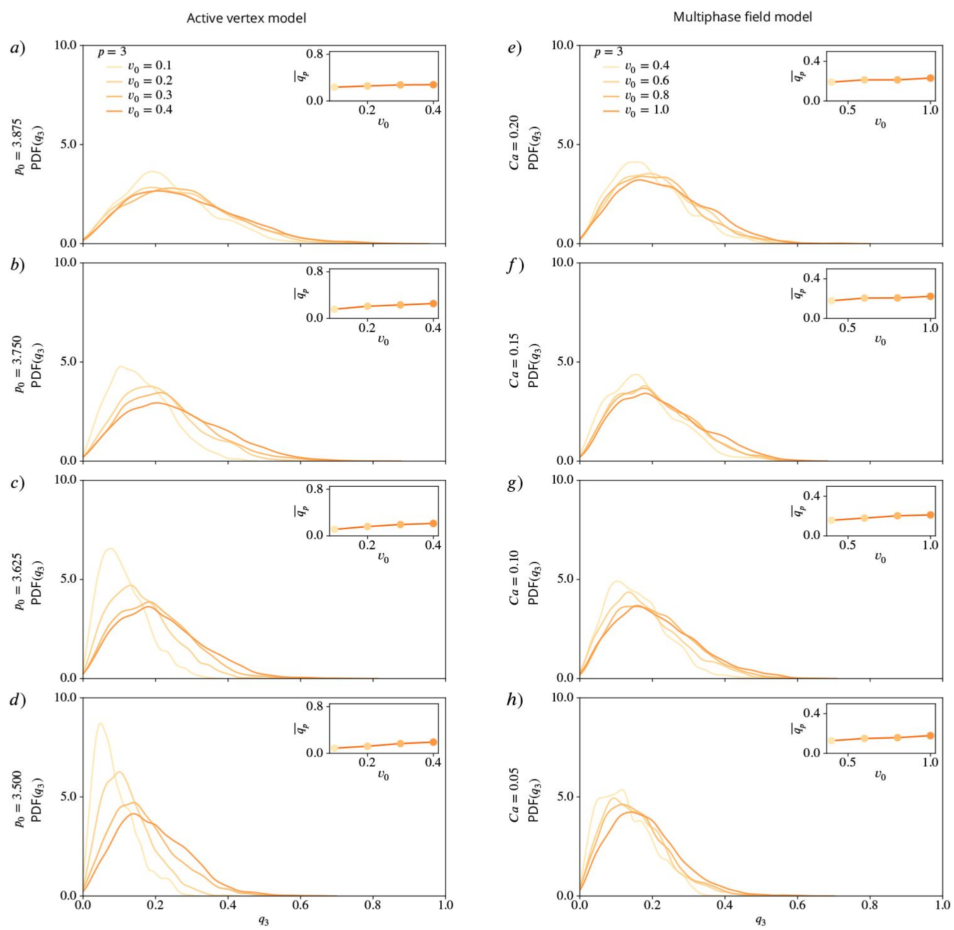
Inlets show mean values of $q_4$ as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.




**Appendix 2—figure 16.**

**PDFs for $q_5$ using kde-plots, for varying deformability $p_0$ or $Ca$ and fixed activity $v_0$.**

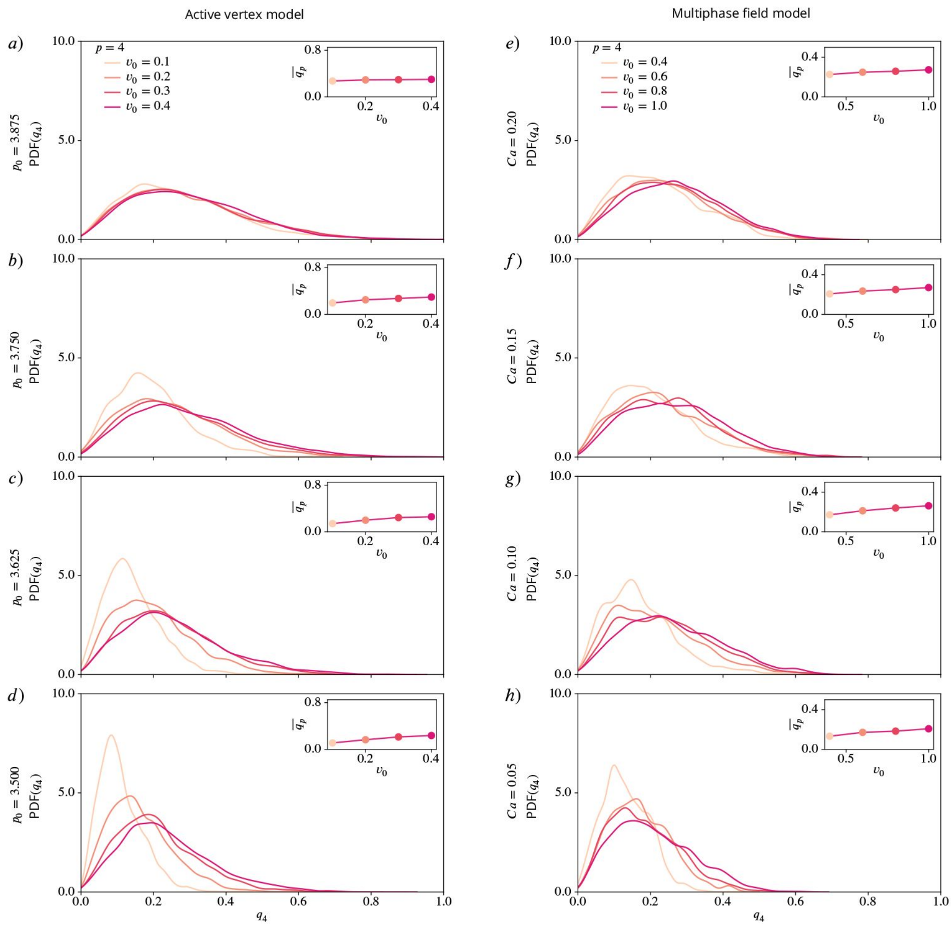
Inlets show mean values of $q_5$ as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.




**Appendix 2—figure 17.**

**PDFs for $q_3$ using kde-plots, for varying activity $v_0$ and fixed deformability $p_0$ or *Ca*.**

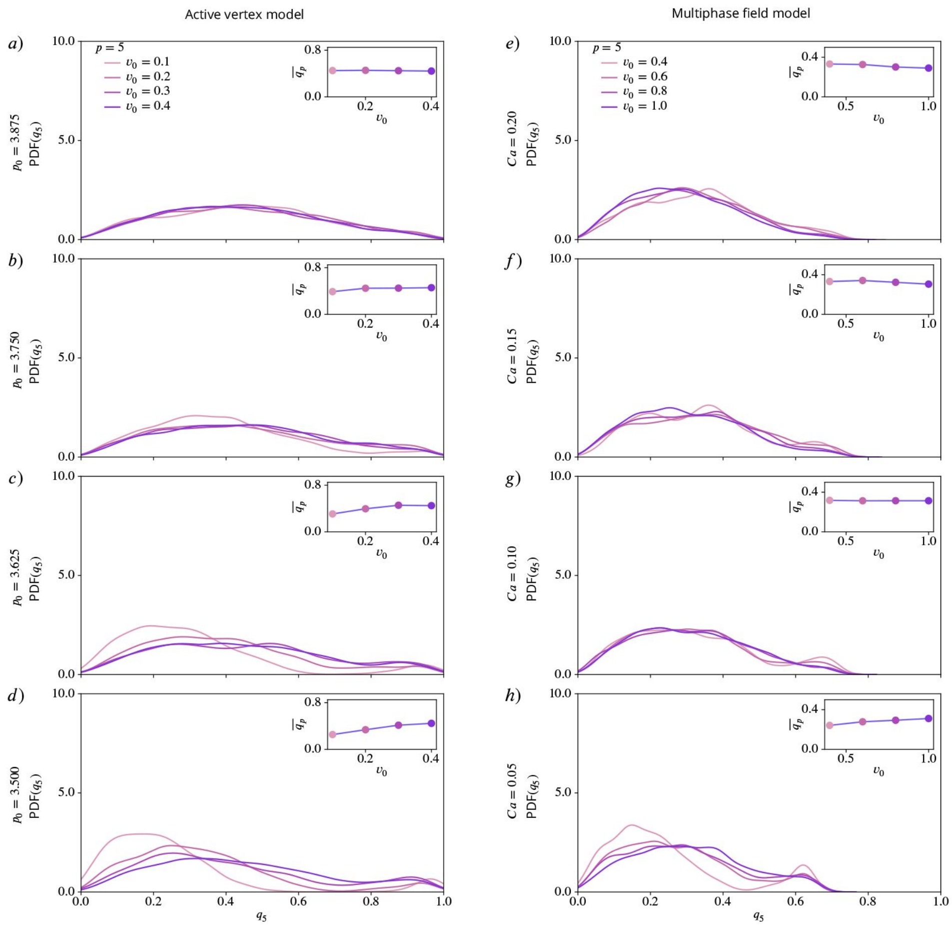
Inlets show mean values of $q_3$ as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.




**Appendix 2—figure 18.**

**PDFs for $q_4$ using kde-plots, for varying activity $v_0$ and fixed deformability $p_0$ or $Ca$.**

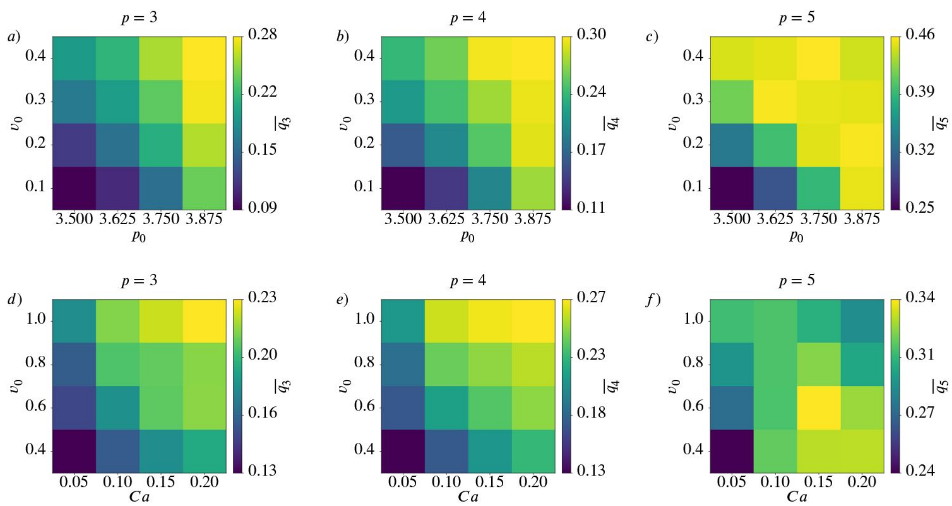
Inlets show mean values of $q_4$ as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.




**Appendix 2—figure 19.**

**PDFs for $q_5$ using kde-plots, for varying activity $v_0$ and fixed deformability $p_0$ or $Ca$.**

Inlets show mean values of $q_5$ as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.




**Appendix 2—figure 20.**

**Mean value $\overline{q_p}$ for *p* = 3 (left), *p* = 4 (middle) and *p* = 5 (right) as function of deformability $p_0$ or *Ca* and activity $v_0$ for active vertex model (a) - c)) and multiphase field model (d) - f)).**

# Appendix 3

## Results using polygonal shape analysis

To visualize the similarities and differences between $q_p$ and $\gamma_p$ we show the analogue of **Figure 4** using the Minkowski tensors in **Appendix 3 Figure 21** using polygonal shape analysis. For regular polygons, both shape characterization methods behave in the same way as seen in the first row. For irregular shapes, like in the second row, they already show a different behaviour. Note that the rounded shapes from **Figure 4** are missing in **Appendix 3 Figure 21** as $\gamma_p$ requires a polygon.

The influence of the choice of the shape characterization method is not only visible in the values for single shapes, it can also be seen in the mean and standard deviation. To illustrate this, we use $\gamma_p$ to characterize the shapes from the same experimental data as in **Figure 1** and **Figure 2**. The segmented data consists of smooth contours for each cell. As $\gamma_p$ only works with vertex coordinates the first step is to identify these. Note that this always leads to the approximation of the cell shape with a polygon and therefore to a simplification of the shape. For finding the vertices we consider the Voronoi interface method (Saye and Sethian, 2011***, 2012***). Using the so obtained vertices and Equation 8 and Equation 9 leads to the results shown in **Appendix 3 Figure 22** and **Appendix 3 Figure 23**. Compared to **Figure 1** and **Figure 2**, where the Minkowski tensor was used for the same data, we see that the choice of the shape characterization method highly influences the results.

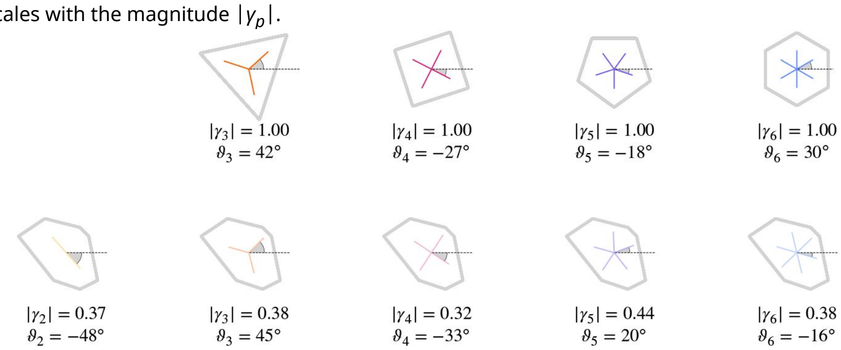
Qualitative consequences of the approach become apparent by comparing **Figure 8** with Appendix 3 Figure 12, which, instead of the monotonic trend for activity and deformability, exhibit nonmonotonic trends, peaking at intermediate values of activity or deformability.

Considering the independence of $\gamma_2$ and $\gamma_6$ the corresponding results to **Figure 7** and the corresponding distance correlation and statistical tests **Figure 7—figure Supplement 1** and **Figure 7— figure Supplement 2**, respectively, are provided in **Appendix 3 Figure 24** and **Appendix 3 Figure 24— figure Supplement 1** and **Appendix 3 Figure 24—figure Supplement 24**, respectively, but only for the active vertex model.

## Appendix 3—figure 21.

**Regular and irregular shapes, adapted from Armengol-Collado et al. (2023), with magnitude and orientation calculated by Equation 8 and Equation 9.**

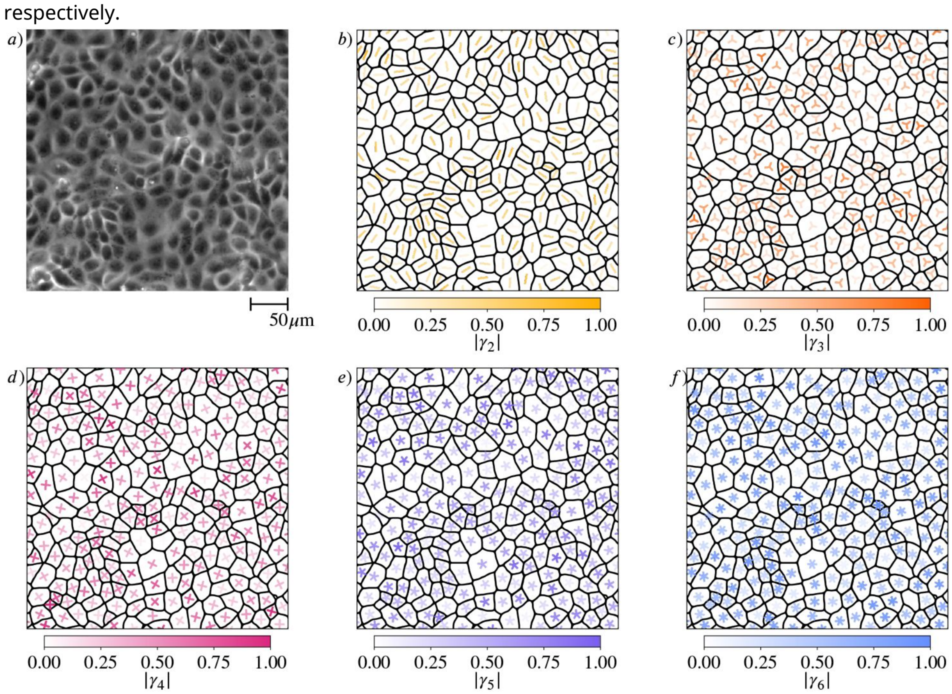
The brightness scales with the magnitude $|\gamma_p|$.

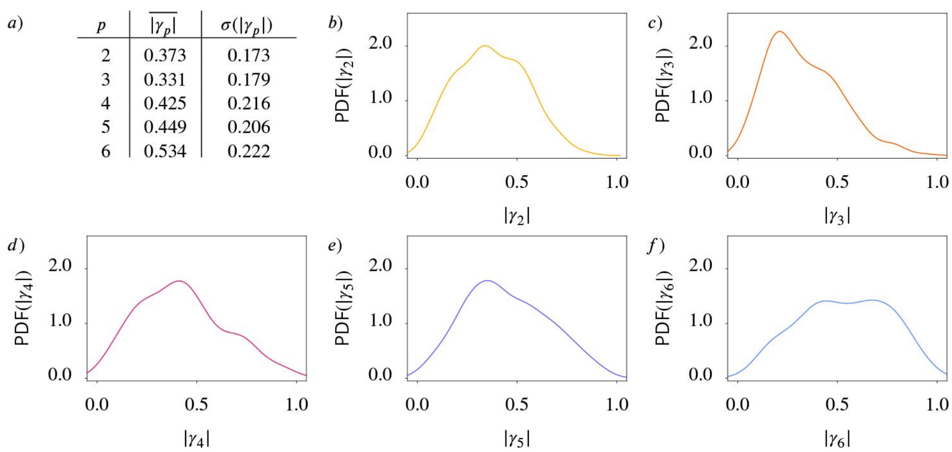



## Appendix 3—figure 22.

**Shape classification of cells in wild-type MDCK cell monolayer.**

a) Raw experimental data. b) - f) Polygonal shape classification, visualized using $\gamma_p$ calculated by **Equation 8** and **Equation 9** for $p$ = 2, 3, 4, 5, 6, respectively. The brightness and the rotation of the $p$-atic director indicates the magnitude and the orientation, respectively.



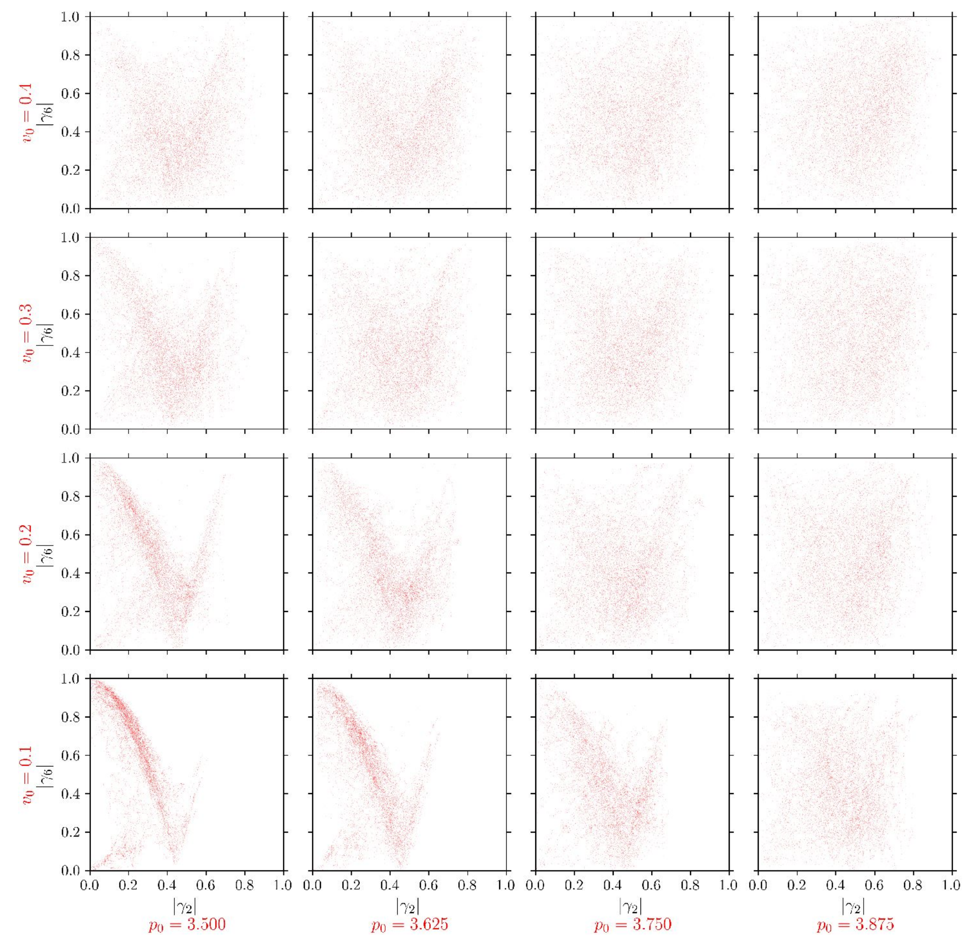



| $p$ | $\overline{\lvert\gamma_p\rvert}$ | $\sigma(\lvert\gamma_p\rvert)$ |
|---|---|---|
| 2 | 0.373 | 0.173 |
| 3 | 0.331 | 0.179 |
| 4 | 0.425 | 0.216 |
| 5 | 0.449 | 0.206 |
| 6 | 0.534 | 0.222 |

**Appendix 3—figure 23.**

**Statistical data for cell shapes identified in Appendix 3 Figure 22.**

(a) Mean $\overline{|\gamma_p|}$ and standard deviation $\sigma(|\gamma_p|)$ of $|\gamma_p|$. b) - f) Probability distribution function (PDF) of $|\gamma_p|$ for $p$ = 2, 3, 4, 5, 6, respectively. Kde-plots are used to show the probability distribution. For this first analysis we regard only one frame with 235 cells.




**Appendix 3—figure 24.**

### $|\gamma_6|$ (y-axis) versus $|\gamma_2|$ (x-axis) for all cells in the active vertex model.

For each cell and each timestep we plot one point ($|\gamma_2|$, $|\gamma_6|$). Each panel corresponds to specific model parameters $p_0$ and $v_0$, representing deformability and activity.

**Appendix 3—figure 24—figure supplement 1**. Distance correlation $dCor(|\gamma_2|, |\gamma_6|)$

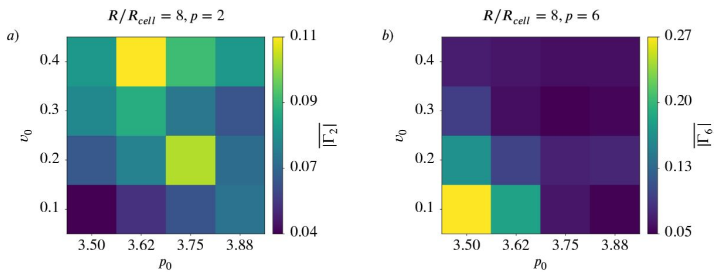
**Appendix 3—figure 24—figure supplement 2**. P-values of the distance correlation $P_{dCor(|\gamma_2|,|\gamma_6|)}$

## Appendix 3—figure 25.

### Coarse-Grained nematic ($p$ = 2) and hexatic ($p$ = 6) order for $R/R_{cell}$ = 8 depending on activity and deformability of the cells.

$\overline{|\Gamma_p|}$ as function of deformability $p$ and activity $v$ for active vertex model. a) nematic order ($p$ = 2), b) hexatic order ($p$ = 6).

**Appendix 3—figure 25—figure supplement 1**. $\overline{\Gamma_6}$ versus $\overline{\Gamma_2}$ for different coarse graining radii in the active vertex model.



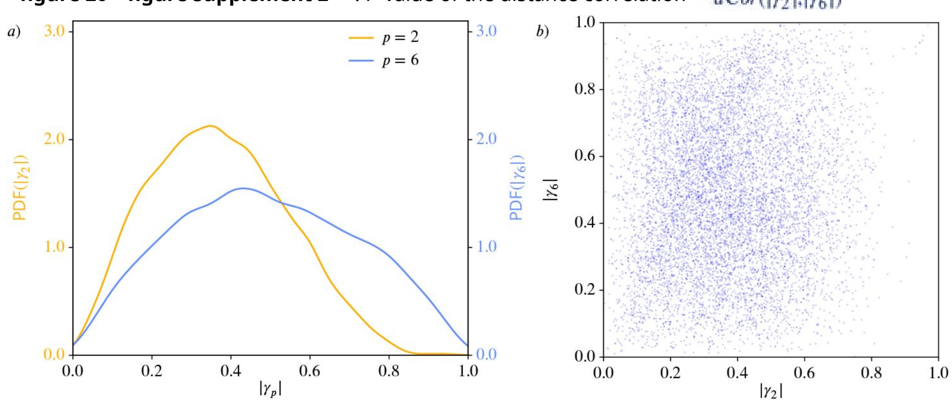

## Appendix 3—figure 26.

### Nematic ($p$ = 2) and hexatic ($p$ = 6) order for the cells in the experiments from Armengol-Collado et al. (2023).

We use the polygonal approximation of the cell shape as $\gamma_p$ can only work with polygons. *a*): Probability distribution functions (PDFs) using kde-plots, for $|\gamma_2|$ (orange) and $|\gamma_6|$ (blue). *b*): Nematic ($p$ = 2) and hexatic ($p$ = 6) order are independent of eachother. $|\gamma_6|$ (y-axis) versus $|\gamma_2|$ (x-axis) for all cells. For each cell and each timestep we plot one point ($|\gamma_2|$, $|\gamma_6|$).

**Appendix 3—figure 26—figure supplement 1**. Distance correlation $dCor(|\gamma_2|, |\gamma_6|)$

**Appendix 3—figure 26—figure supplement 2**. P-value of the distance correlation $P_{dCor(|\gamma_2|,|\gamma_6|)}$



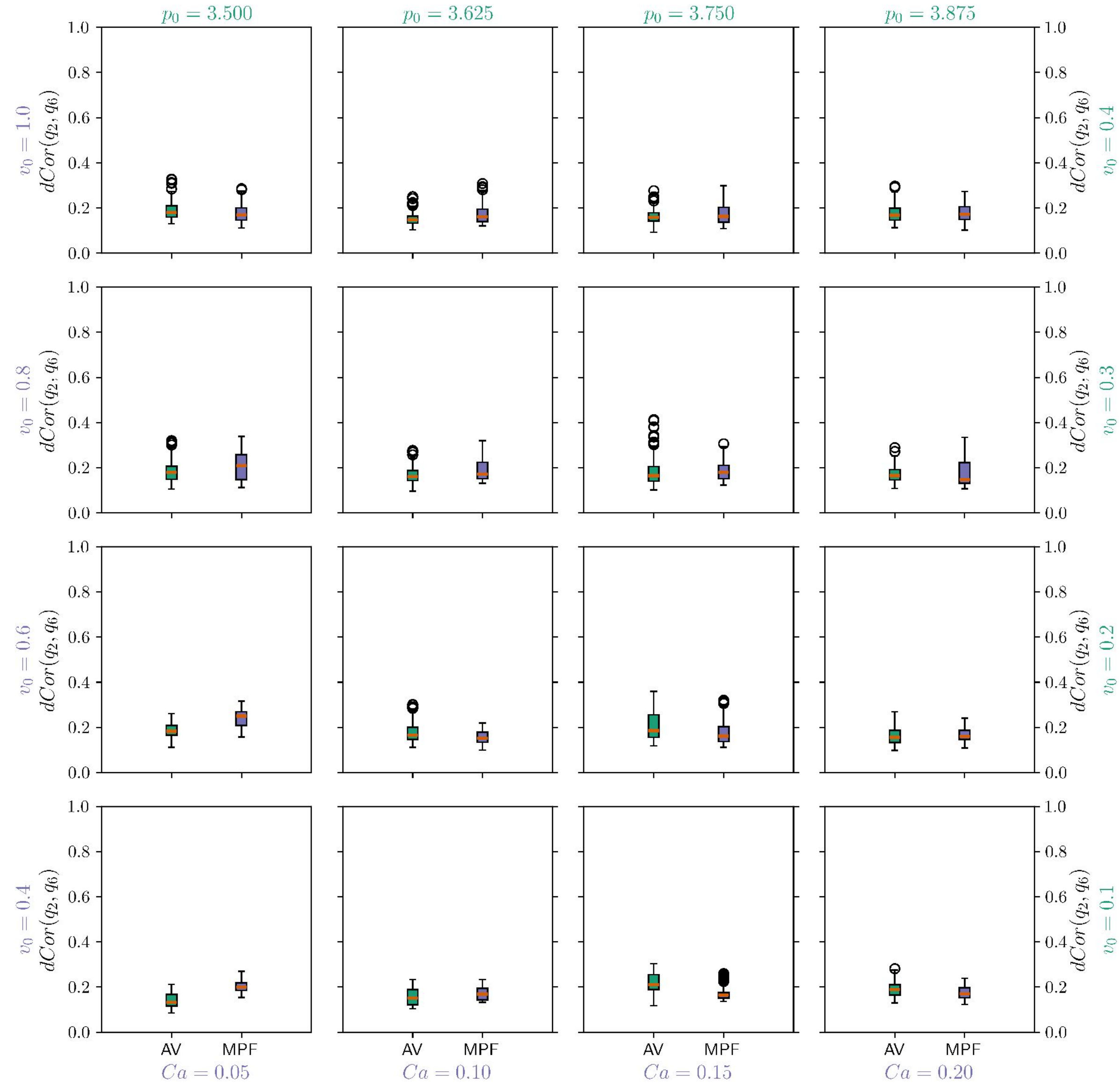



**Figure 7—figure supplement 1.**

**Distance correlation *dCor*($q_2$, *q*6) between $q_2$ and $q_6$ for all cells in the multiphase field model (MPF - purple box) and active vertex model (AV - green box).**

Each panel corresponds to specific model parameters; *Ca* and $v_0$ for multiphase field model, and $p_0$ and $v_0$ for the active vertex model, representing deformability and activity, respectively. We compute one value for every timestep and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 × (Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles.

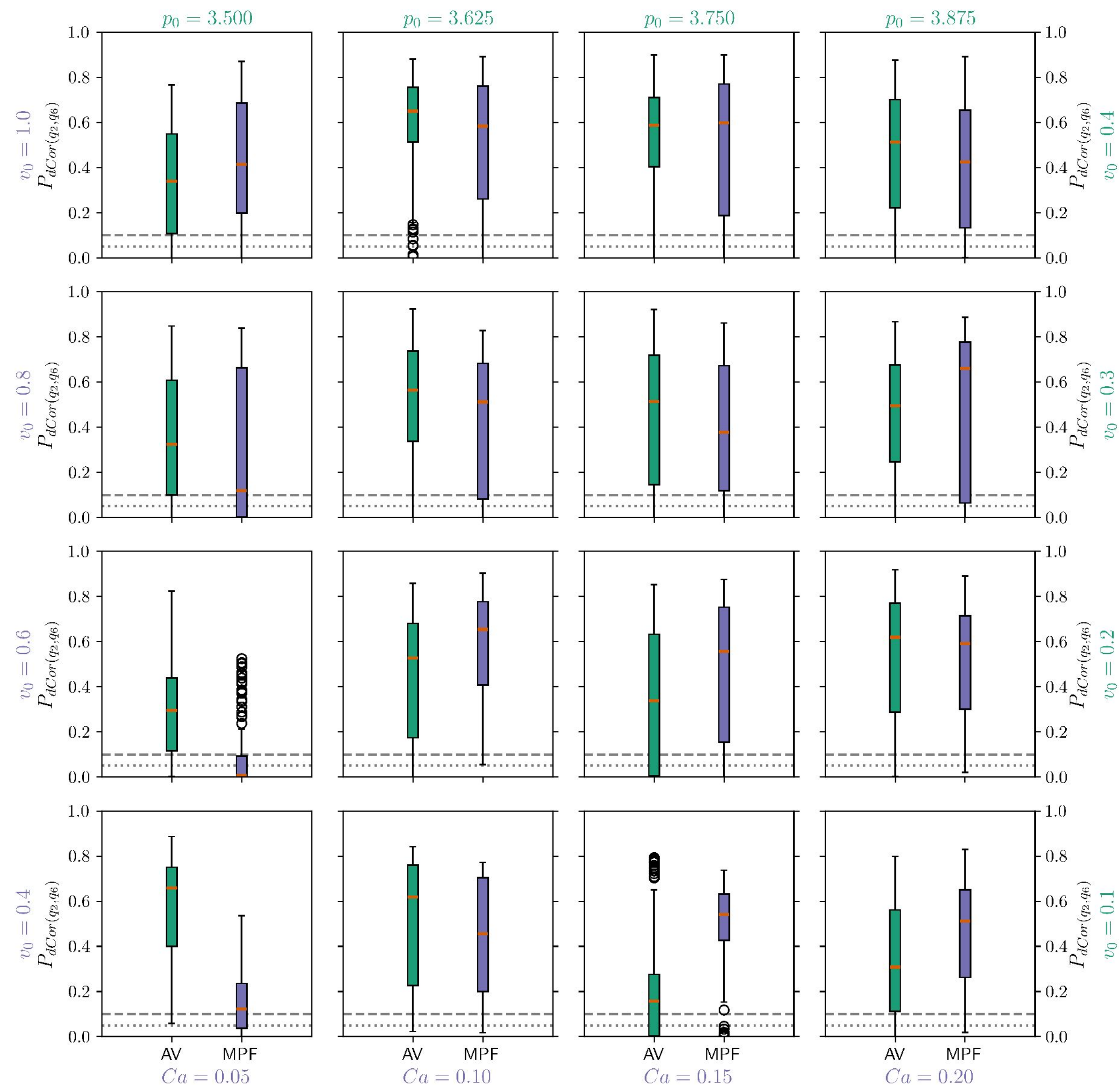



**Figure 7—figure supplement 2.**

**P-values of the distance correlation $P_{dCor(q_2,q6)}$ between $q_2$ and $q_6$ for all cells in the multiphase field model (MPF - purple box) and active vertex model (AV - green box).**

Each panel corresponds to specific model parameters; *Ca* and $v_0$ for multiphase field model, and $p_0$ and $v_0$ for the active vertex model, representing deformability and activity, respectively. We compute one value for every timestep and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 ×(Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles. 0.05 and 0.1 are marked with a grey dotted/grey dashed line to guide the eye.

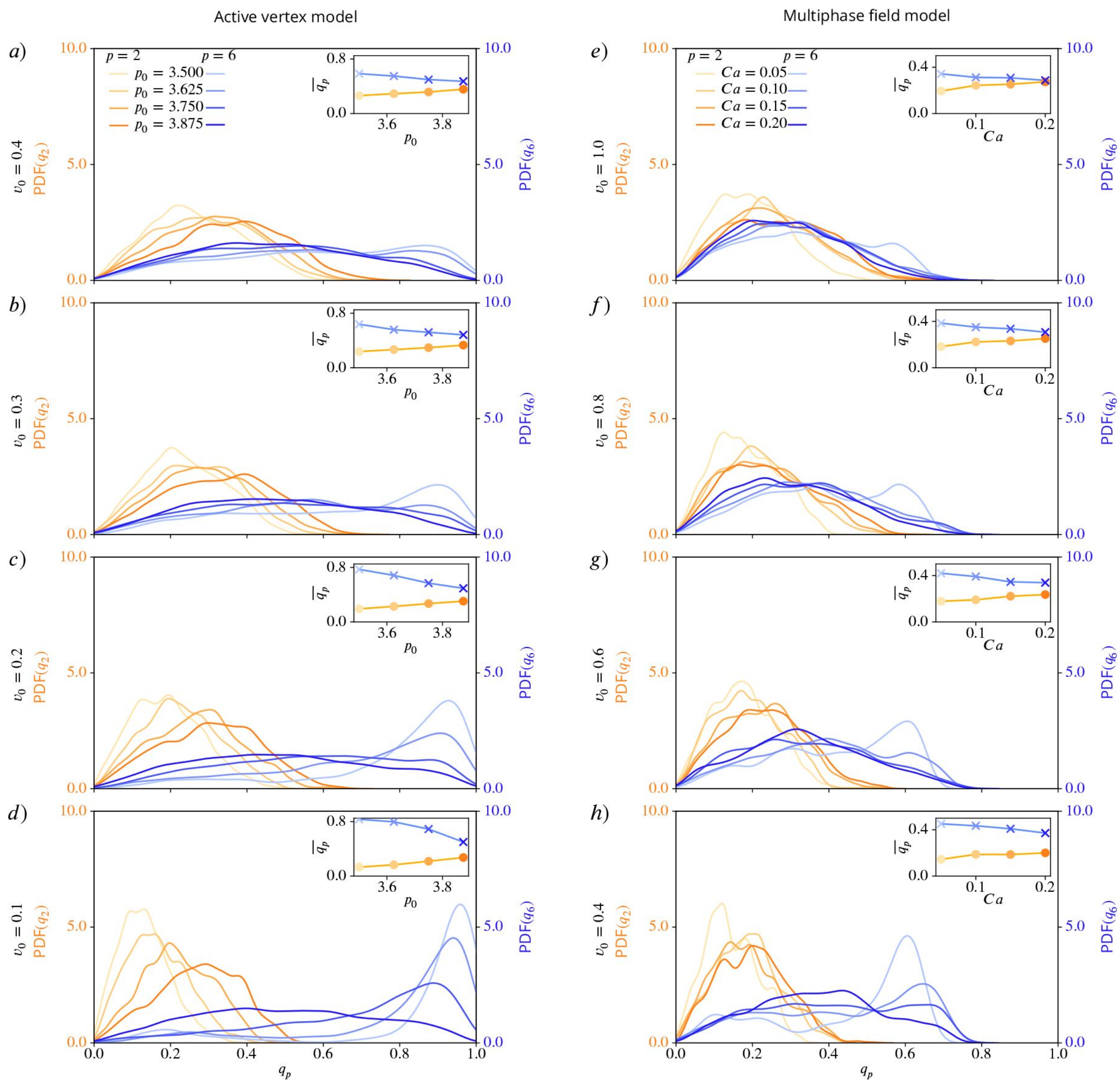



**Figure 8—figure supplement 1.**

**Nematic (*p* = 2) and hexatic (*p* = 6) order depend on deformability of the cells.**

Probability distribution functions (PDFs) for $q_2$ (shades of orange) and $q_6$ (shades of blue), using kde-plots, for varying deformability $p_0$ or $Ca$ and fixed activity $v_0$. Inlets show mean values of $q_2$ and $q_6$ as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.

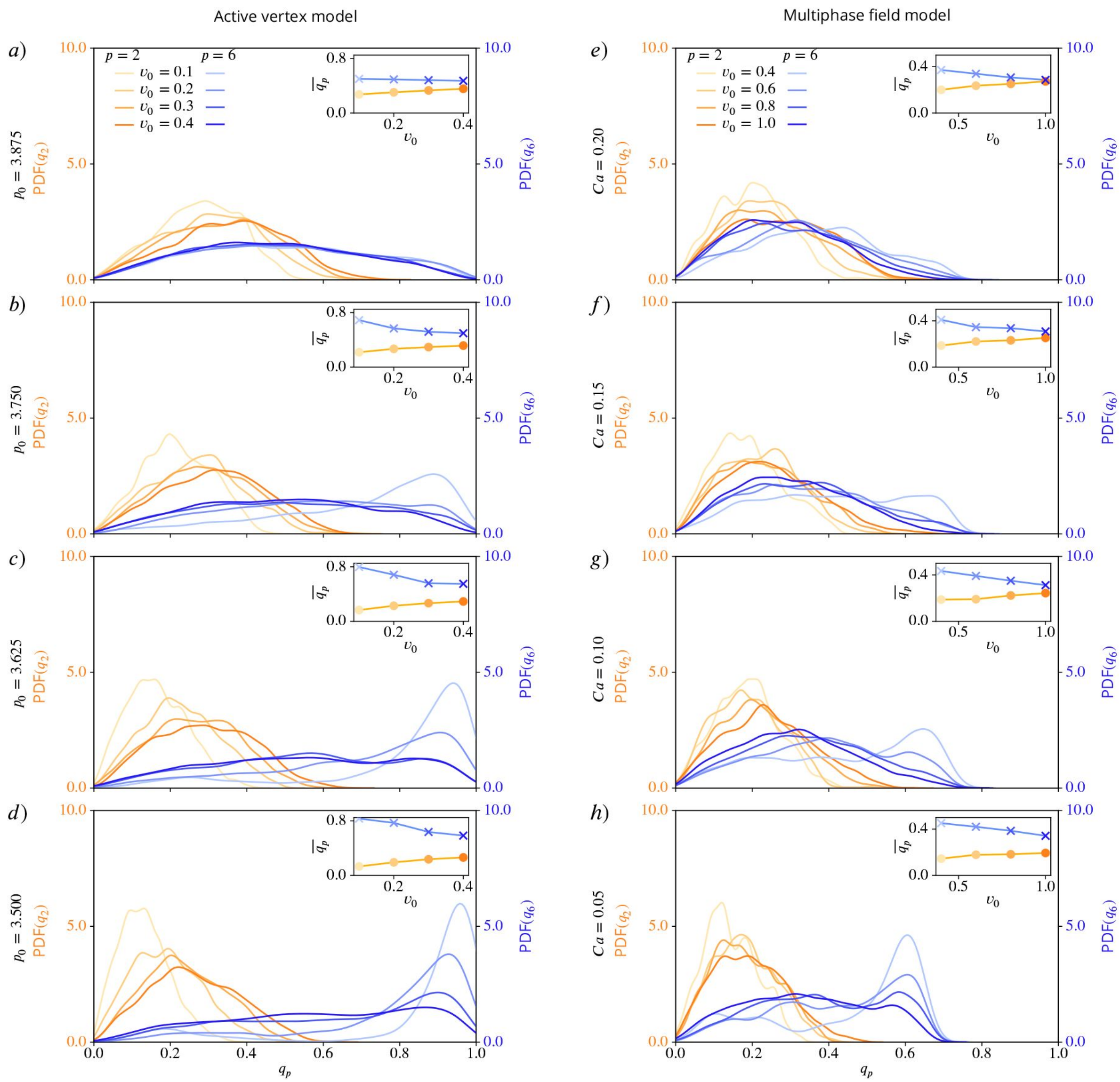



**Figure 8—figure supplement 2.**

## Nematic ($p$ = 2) and hexatic ($p$ = 6) order depend on activity of the cells.

Probability distribution functions (PDFs) for $q_2$ (shades of orange) and $q_6$ (shades of blue), using kde-plots, for varying activity $v_0$ and fixed deformability $p_0$ or $Ca$. Inlets show mean values of $q_2$ and $q_6$ as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.

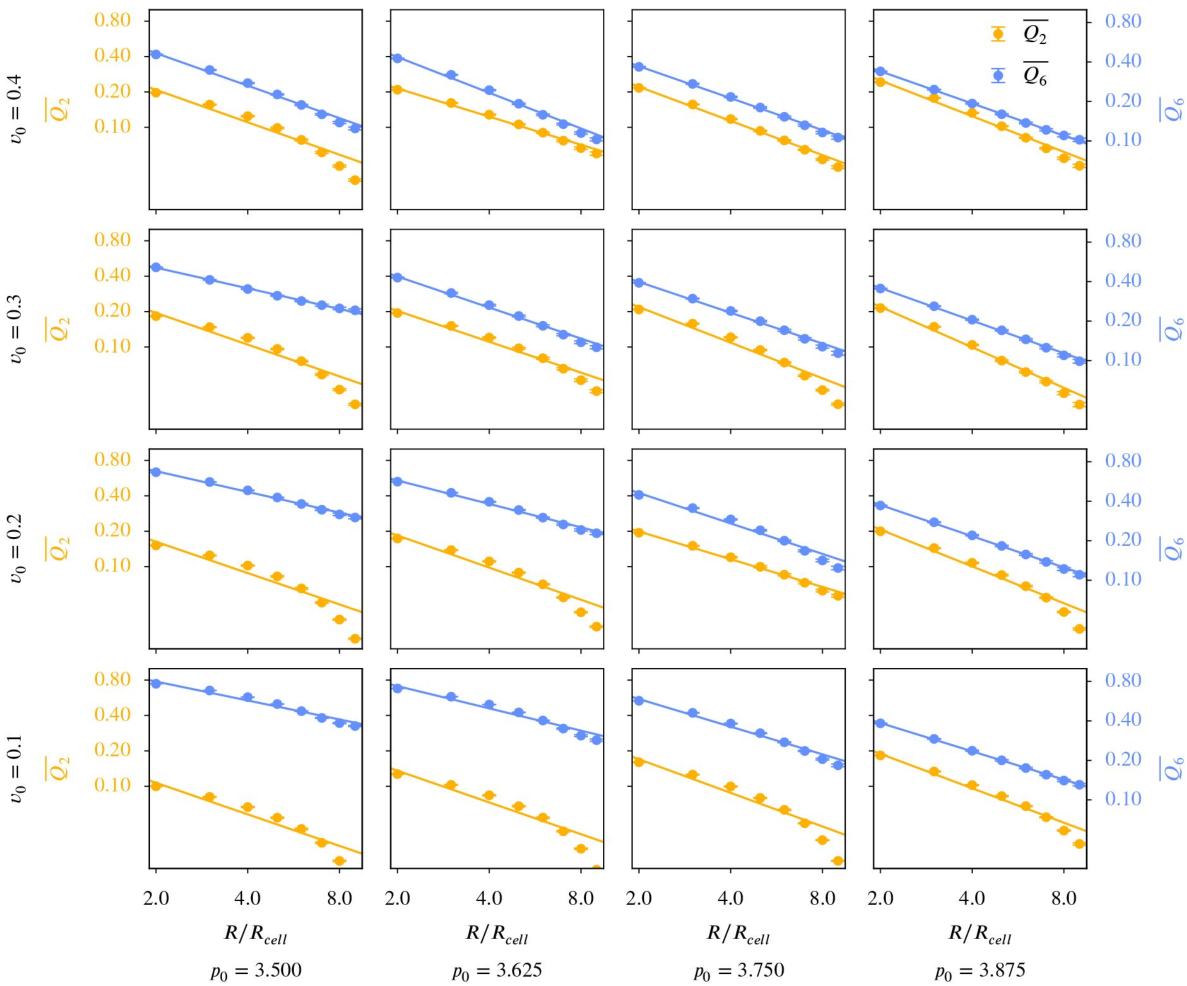



**Figure 9—figure supplement 1.**

## $\overline{Q_6}$ versus $\overline{Q_2}$ for different coarse graining radii in the active vertex model.

$Q_p$ was calculated according to **Equation 11**, the averaging of this and the choice of $R_{\mathscr{C}}$ follow the description in Coarse-grained quantities. The maximal coarse-graining radius corresponds to half the domain width. Each panel corresponds to specific model parameters; $p_0$ and $v_0$, representing deformability and activity. A logarithmic scaling was used for both axis. Error bars are obtained as s.e.m..

**Figure 9—figure supplement 2.**

### $\overline{Q_6}$ versus $\overline{Q_2}$ for different coarse graining radii in the multiphase field model.

$Q_p$ was calculated according to **Equation 11**, the averaging of this and the choice of $R_{\mathscr{C}}$ follow the description in Coarse-grained quantities. The maximal coarse-graining radius corresponds to half the domain width. Each panel corresponds to specific model parameters; *Ca* and $v_0$, representing deformability and activity. A logarithmic scaling was used for both axis. Error bars are obtained as s.e.m..

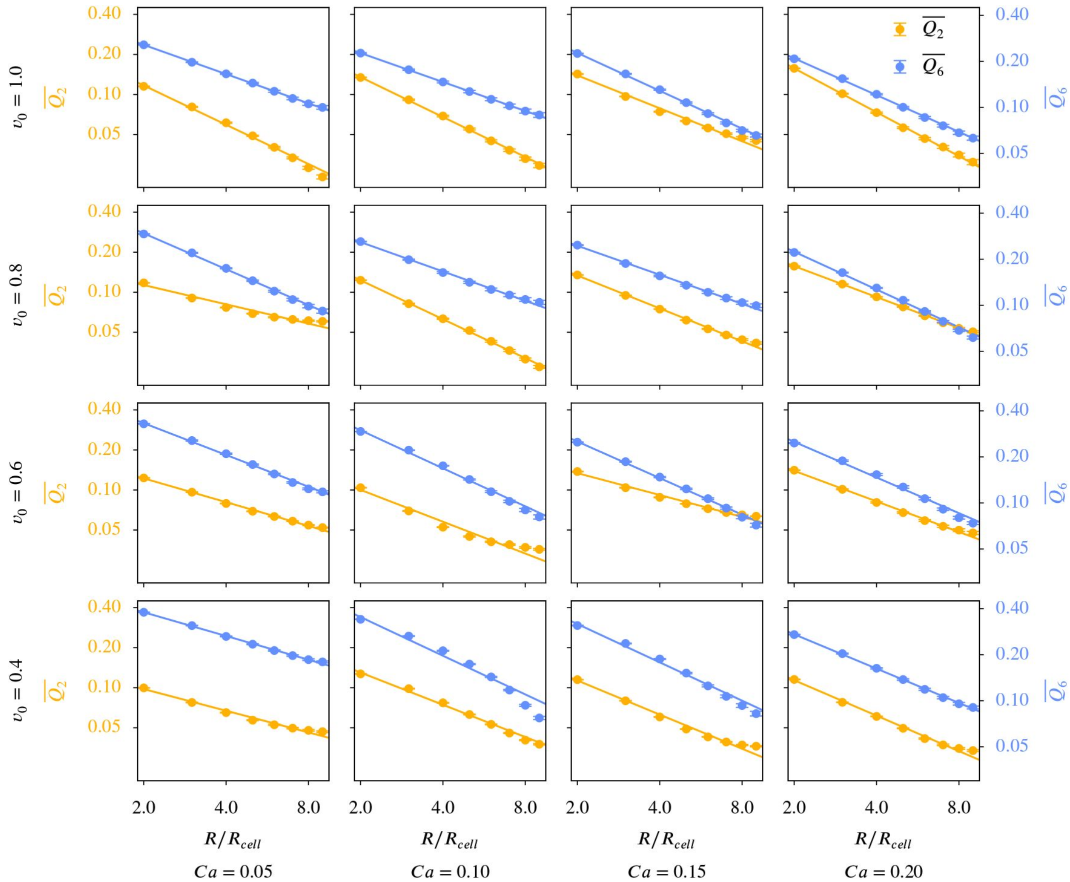
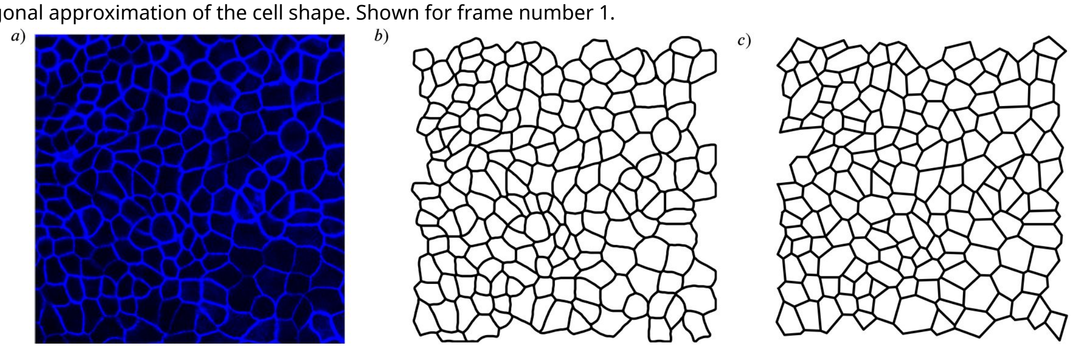



**Figure 10—figure supplement 1.**

### Cell shapes in the experiments from Armengol-Collado et al. (2023).

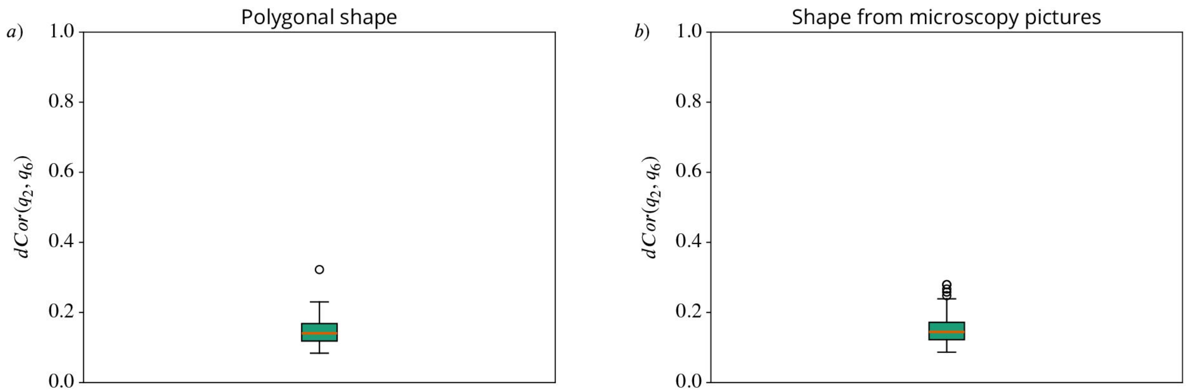
a) and the cell shapes used for the shape quantification, b) detailed cell outline obtained from the microscopy image, c) polygonal approximation of the cell shape. Shown for frame number 1.



## Figure 10—figure supplement 2.

### Distance correlation $dCor(q_2, q6)$ between $q_2$ and $q_6$.

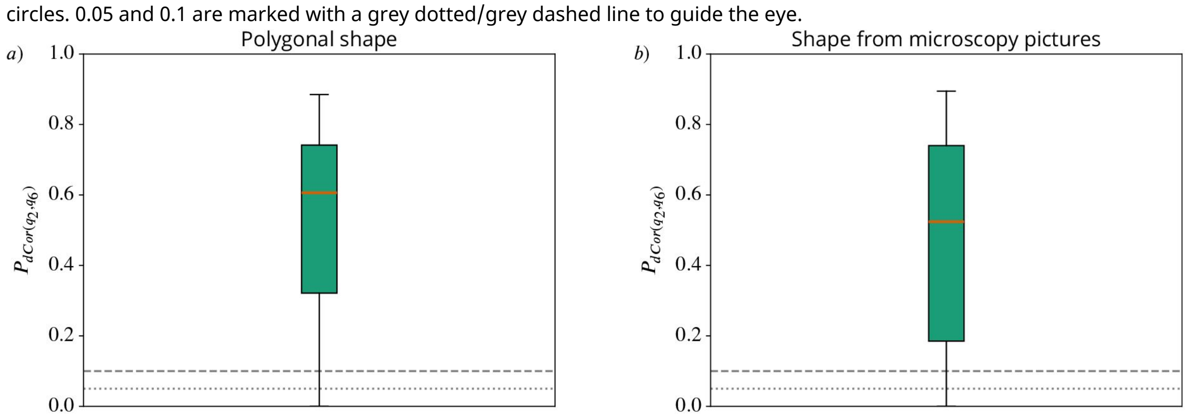
On the left side *a*) we use the polygonal approximation of the cell shape, on the right side *b*) we use the detailed cell outline obtained from the microscopy pictures. We compute one value for every frame and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 × (Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles.

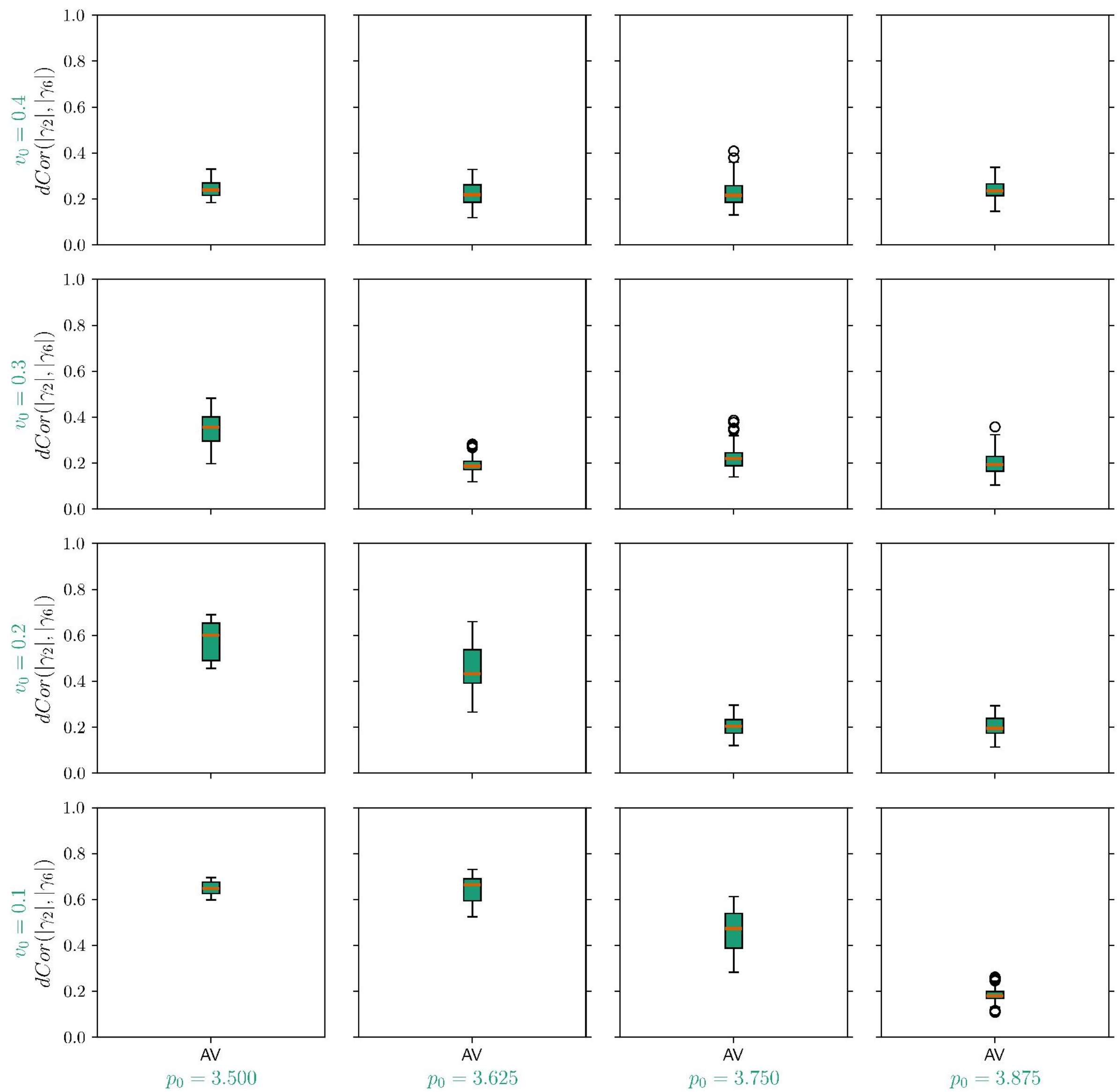



## Figure 10—figure supplement 3.

### P-values of the distance correlation $P_{dCor(q_2,q6)}$ between $q_2$ and $q_6$.

On the left side *a*) we use the polygonal approximation of the cell shape, on the right side *b*) we use the detailed cell outline obtained from the microscopy pictures. We compute one value for every frame and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 × (Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles. 0.05 and 0.1 are marked with a grey dotted/grey dashed line to guide the eye.

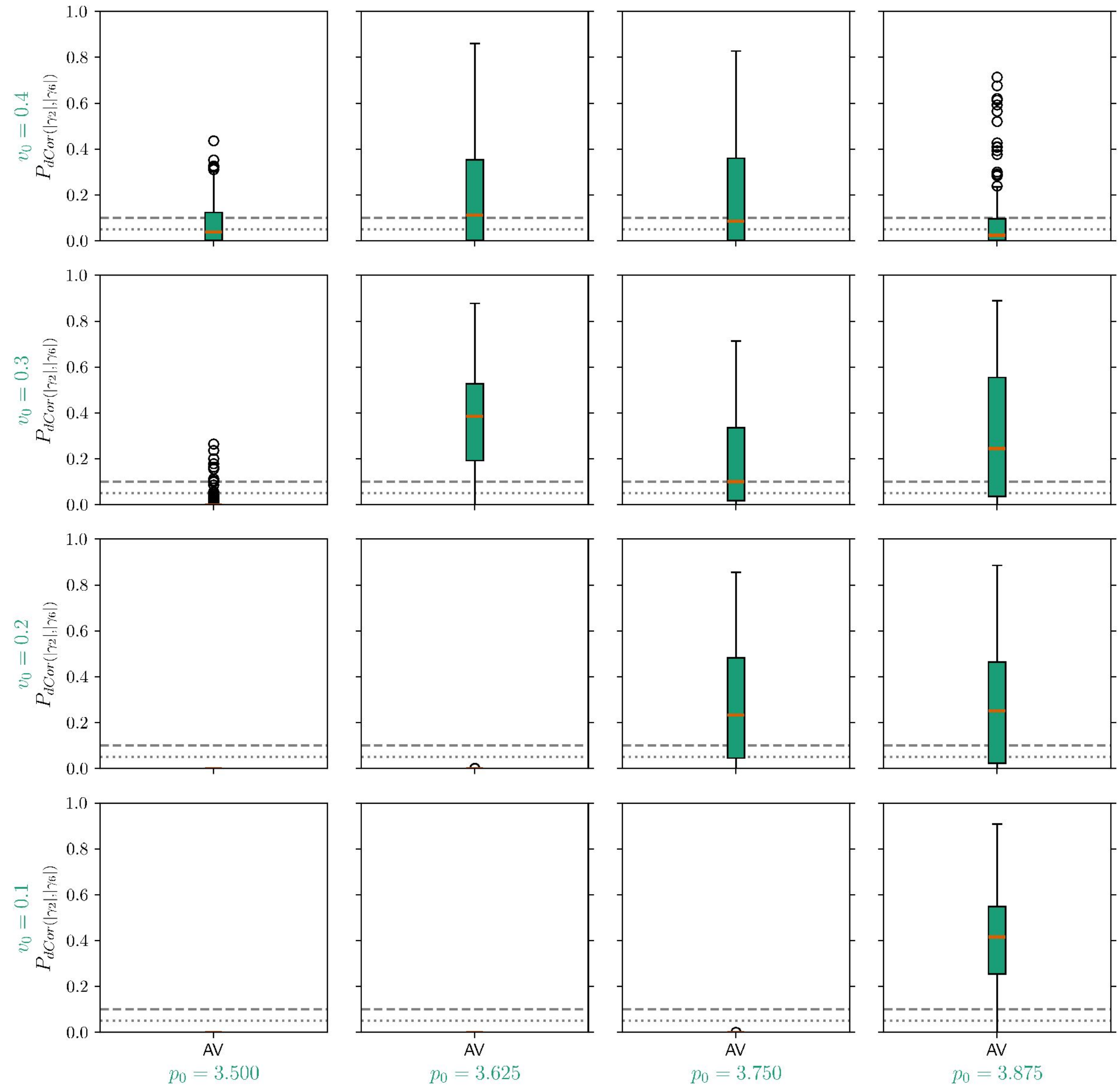





**Appendix 3—figure 24—figure supplement 1.**

### Distance correlation $dCor(|\gamma_2|, |\gamma_6|)$ for the simulation data of the active vertex model.

Each panel corresponds to specific model parameters $p_0$ and $v_0$, representing deformability and activity. We compute one value for every timestep and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 × (Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles.




**Appendix 3—figure 24—figure supplement 2.**

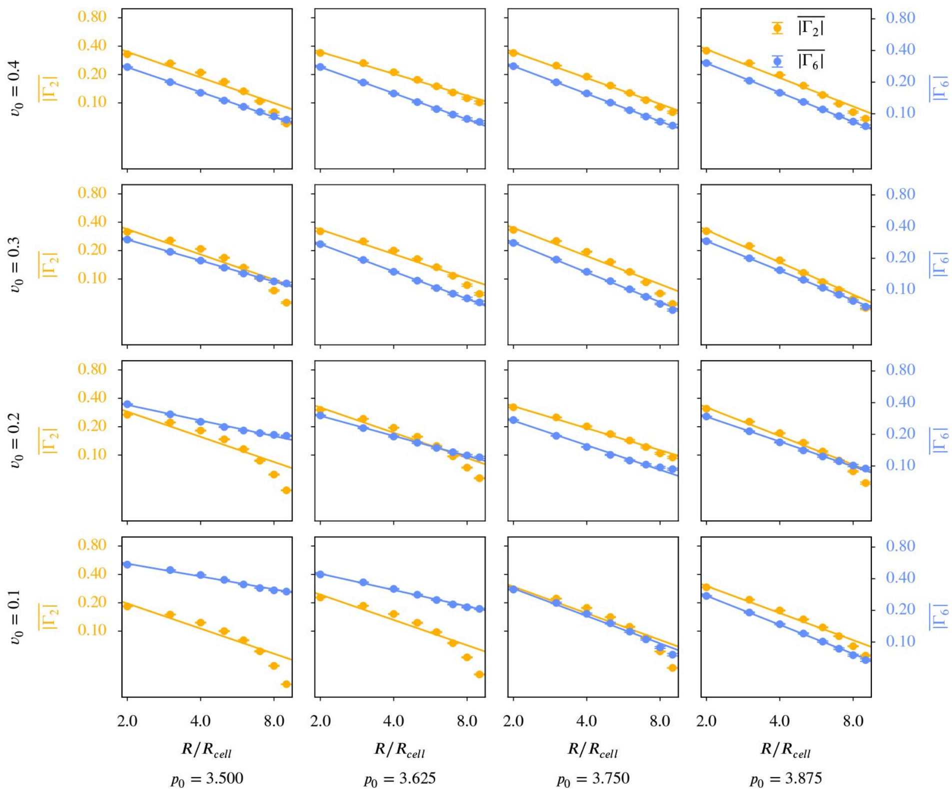
**P-values of the distance correlation $P_{dCor(|\gamma_2|,|\gamma_6|)}$ for the simulation data of the active vertex model.**

Each panel corresponds to specific model parameters $p_0$ and $v_0$, representing deformability and activity. We compute one value for every timestep and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 × (Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles. 0.05 and 0.1 are marked with a grey dotted/grey dashed line to guide the eye.

## Appendix 3—figure 25—figure supplement 1.

**$\overline{\Gamma_6}$ versus $\overline{\Gamma_2}$ for different coarse graining radii in the active vertex model.**

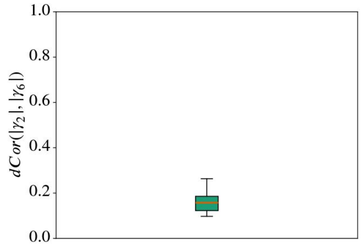
$\Gamma_p$ was calculated according to Equation 12, the averaging of this and the choice of $R_{\mathscr{C}}$ follow the description in Coarse-grained quantities. The maximal coarse-graining radius corresponds to half the domain width. Each panel corresponds to specific model parameters; $p_0$ and $v_0$, representing deformability and activity. A logarithmic scaling was used for both axis. Error bars are obtained as s.e.m..

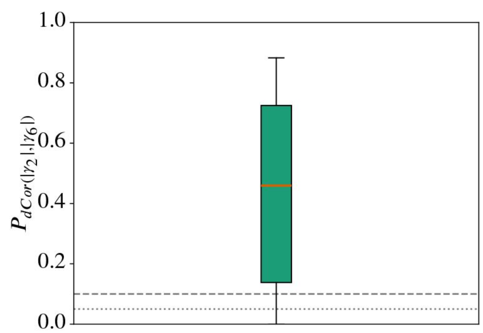



## Appendix 3—figure 26—figure supplement 1.

**Distance correlation $dCor(|\gamma_2|, |\gamma_6|)$ for all cells from the experimental data in Armengol-Collado et al. (2023).**

We use the polygonal approximation of the cell shape as $\gamma_p$ can only work with polygons. We compute one value for every frame and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 × (Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles.






**Appendix 3—figure 26—figure supplement 2.**

**P-value of the distance correlation $P_{dCor(|\gamma_2|,|\gamma_6|)}$ for all cells from the experimental data in Armengol-Collado et al. (2023).**

We use the polygonal approximation of the cell shape as $\gamma_p$ can only work with polygons. We compute one value for every frame and present the resulting distribution with a box plot. In the box plots the orange line corresponds to the median of the data and the box ranges from the first to the third quartile of the data. The whiskers go from the lowest data point greater than (value of the first quartile) − 1.5 × (Interquartile range) to the highest data point below (value of the third quartile) + 1.5 × (Interquartile range). Outliers are shown with circles.

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## Author information

**Lea Happel**[†]
Institute of Scientific Computing, TU Dresden, Dresden, Germany

[†]These authors contributed equally to this work

**Griseldis Oberschelp**[†]
Institute of Scientific Computing, TU Dresden, Dresden, Germany

[†]These authors contributed equally to this work

**Valeriia Grudtsyna**
Niels Bohr Institute, University of Copenhagen, Kobenhavn, Denmark

**Harish P Jain**
Njord Centre, Physics Department, University of Oslo, Oslo, Norway

**Rastko Sknepnek**
School of Life Sciences, University of Dundee, Dundee, United Kingdom, School of Science and Engineering, University of Dundee, Dundee, United Kingdom
ORCID iD: 0000-0002-0144-9921

**Amin Doostmohammadi**
Niels Bohr Institute, University of Copenhagen, Kobenhavn, Denmark
ORCID iD: 0000-0002-1116-4268

**Axel Voigt**
Institute of Scientific Computing, TU Dresden, Dresden, Germany, Center of Systems Biology Dresden, Dresden, Germany, Cluster of Excellence, Physics of Life, TU Dresden, Dresden, Germany
ORCID iD: 0000-0003-2564-3697

**For correspondence:** axel.voigt@tu-dresden.de

## Editors

Reviewing Editor
**Karsten Kruse**
University of Geneva, Geneva, Switzerland

Senior Editor
**Aleksandra Walczak**
CNRS, Paris, France

**Reviewer #1 (Public review):**

Summary:

The authors stated aim is to introduce so-called Minkowski tensors to characterize and quantify the shape of cells in tissues. The authors introduce Minkowski tensors and then define the p-atic order q_p as a cell shape measure, where p is an integer. They also introduce a previously defined measure of p-atic order in the form of the parameter \gamma_p. The authors compute q_p for data obtained by simulating an active vertex model and a multiphase field model, where they focus on p=2 and p=6 - so-called nematic and hexatic order - as the two values of highest biological relevance. Based on their analysis, the authors state that q_2 and q_6 are independent, that there is no crossover for the coarse-grained quantities, that a comparison of q_p for different values of p is not meaningful, and determine the dependence of the mean value of q_2 and q_6 on cell activity and deformability. Subsequently, they apply their method to data from MDCK monolayers and argue that the full range of q_p values needs to be considered to characterize shape and positional order in epithelia..

Strength:

The work presents a set of parameters that are useful for analyzing cell shape.

Weaknesses:

The introduction of the Minkowski tensors is hardly accessible for typical biologists. Eventually, most quantification is done using q_p, which can be defined without recursion to Minkowski functionals. The relation to Minkowski functionals makes the important properties of robustness and stability evident. However, for an audience of biologists, the derivation of this property could be relegated to an Appendix. Instead, the text could directly go to the results of the analysis of experimental and modeling data.

Important details about how the cell shapes are extracted from the experimental data are missing. The two data sets the authors consider are not analyzed in the same way.

https://doi.org/10.7554/eLife.105680.2.sa2

**Reviewer #3 (Public review):**

Hapel et al. present an article entitled Quantifying the shape of cells - from Minkowski tensors to p-atic order. The paper reports the p-atic quantitative method - established in physics - to extract cell full shapes in biological experiments using their images of epithelial MDCK cells (phase contrast) and also images reported in another paper as well as their own simulations based on active vertex model and multiphase phase fields approaches. Authors present the rationale of this new strategy for quantification. They adapt the method of Minkowski tensors and they extract distributions of cell shapes readouts with plots of their distributions. An emphasis is given to changes in cell shapes captured by this method. Higher rank tensors are considered as well as representations with intuitive meanings and q_i orders and their potential correlations or absence of correlations - for example q_2 and q_6, leading to statements about nematic and hexatic orders.

This analysis and its strength are contrasted with Armengol-Collade et al. (2023) quoted in the paper, who consider polygonal shapes for cells and their shape function $\gamma$_p. Authors support the notion of a key improvement thanks to Minkowski tensors approach and doing so, they challenge the former crossovers correlations statements reported in Armengol-Collade et al. (2023). In this context, they defend that nematic liquid crystals approach is not sufficient to capture cell dynamics in tissues. Also they propose that q_2 and q_6 could serve as readout for activity and deformability of cells among other statements related to their approach.

A variety of analytical methods have been realised to track cells in monolayers in vitro and in vivo during morphogenesis - for example, shear decomposition (from MPI-PKS Dresden) or links joining centroids and their neighbours approach (MSC/Curie Paris) to name few examples. It will be interesting in the future that systematic comparisons between these analytical methods are performed with highlights on their respective advantages and drawbacks. This will allow experimentalists to identify the best relevant methods to address their morphogenetic questions.

https://doi.org/10.7554/eLife.105680.2.sa1

**Author response:**

The following is the authors' response to the original reviews

> ***Reviewer 1:***
>
> *I would suggest that the authors focus on what I think is the main goal of the work, namely, to consider the whole cell contour when characterizing cell shape instead of only some points on the contour. A reference to the connection with Minkowski tensors and the biologically relevant mathematical consequences of this connection would suffice; a detailed definition of the Minkowski tensors does not seem to be necessary. Especially because you do not really use them. You could use the analysis of the simulation data to explain what the $\gamma_p$ miss and for which statements they would be sufficient.*

We argue that the explanation of Minkowski tensors is helpful and should remain in the Methods and materials section. There are two reasons: First, our argumentation relays on the robustness and stability properties of Minkowski tensors. Introducing $q_p$ without the connection to Minkowski tensors would not allow us to make these statements. Second, Minkowski tensors seem not well known in the community, otherwise measures like $\gamma_p$ would not have been introduced. Furthermore, readers not interested in the technical details could skip this part of the manuscript and directly go to the Results section. Concerning the questions, what the $\gamma_p$ miss and for which statements they would be sufficient, the answer from a purly mathematical point of view is rather simple: As $\gamma_p$ does not share robustness and stability it should not be used in any case! The provided results on computational and experimental data demonstrate the consequences of using such measures. In case of the proposed nematic-hexatic transition in Armengol-Collade et al. (2023) the consequence is severe, as this transition is specific only to the used method but not to the underlying physics. A second aspect which we now further highlight is the influence of approximating a cell by a polygon. We demonstrate that this approximation is responsible for a strong hexatic order on the cellular scale in the considered MDCK data from Armengol-Collade et al. (2023).

> *It is not clear to me what we should learn about the two tissue models by using $q_2$ and $q_6$ to quantify cell shape. Can you clearly formulate one or more conclusions?*

What we can learn from the research is a dependence of $q_p$ on model parameters in the two tissue models is

– $\overline{q_2}$ increases with higher activity or deformability

– $\overline{q_6}$ decreases with higher activity or deformability.

Furthermore, $q_2$ and $q_6$ are independent and describe distinct properties. Using these models as a basis to coarse-grain and derive continuous models on the tissue scale, these results indicate that more general p-atic liquid crystal theories should be used and the simplest nematic liquid crystal theories might not be sufficient.

> *The experimental data and their analysis does not seem to add anything to the work. Do you report only data from independent measurements, or did you consider all images of a monolayer?*

As we now also analyze experimental data from Armengol-Collado et al. (2023) which confirm our findings on independency of $q_2$ and $q_6$ and also confirm that the proposed nematic-hexatic transition is only specific to the use of $\gamma_p$ for characterizing the shape, additional experimental data are indeed no longer needed. We, therefore, skip the detailed analysis of this data and only keep the results in Fig 1 and Fig 2 and the corresponding figures in the appendix as illustrating examples.

> *L13: "P-atic liquid crystal theories offer new perspectives on how cells self-organize (...)" This is a difficult entry, because the average reader of eLife might not be familiar with p-atic liquid crystals.*

We agree that p-atic liquid crystals might not be familiar to the average reader. For this reason we introduce orientational order in the introduction with examples demonstrating that not only nematic, but also tetratic and hexatic order have been identified in tissue and introduce the different symmetries. Furthermore, we provide examples for p-atic liquid crystals from other fields and various references. In the conclusion, we also cite models for p-atic liquid crystal theories. Even if the average reader is not familiar with these theories, it should become evident that nematic order might not be sufficient to describe tissue as other symmetries are present as well.

> *L32: "nematic" needs to be introduced.*

Nematic order is already explained as rotational order with 180° degrees. The references cited discuss nematic liquid crystals in the context of morphological changes in tissue. We therefore only added a standard text book as reference for liquid crystal theories and refrain introducing it in more detail in the manuscript.

> *Figure 1: Why do you show the data for $q_3$, $q_4$, and $q_5$, which you do not really consider in this manuscript? Same for Figure 2. Why not combine the two figures? Furthermore, you show $q_p$ without having defined them yet.*

We consider all $p = 2,3,4,5,6$, but focus on $p = 2,6$ in the main text and $p = 3,4,5$ in the appendix. Figures 1 and 2 essentially only introduce the subject and help to relate p-atic order to cell shapes and introduce the methodology to analyze the data. Our conclusion is that all p can be important and should be considered in continuous descriptions of tissue.

> *Equation 1: The notation is confusing: the domain of integration ($C$ or $\partial C$) also appears as the variable you integrate.*

The equation is correct. The variable of integration is 1 or $H$ and the domain of integration is $C$ (cell) or $\partial C$ (cell contour).

> *L68: "a snapshot of the considered monolayer of wild-type MDCK cells". Did you analyse only one monolayer? Please, provide information about the number of monolayers that were imaged and how many cell shapes were analyzed.*

We have analyzed one monolayer and have added the missing information.

> *L86: "field-specific prefactors" I do not understand what is meant by these.*

Different communities, e.g. physics, mathematics, cosmology, .... use different prefactors in the definition. We have removed this statement.

> *L89: "Hadwiger's characterization theorem". What is this?*

This mathematical result is important to claim robustness and stability, it can be found in the cited reference.

> *L104: "the essential property is the continuity". Essential for what?*

Essential "for our purpose" to characterize the shape of cells by a robust method.

> *L120: "the theory also guarantees robust description of p-atic orientation for* $p = 3,4,5,6,...$*" I do not understand what you mean.*

The previous examples only consider $p = 2$. However, the cited theoretical results also hold for $p = 3,4,5,6,..$

> *Equations (5) and (6): You define $\psi_p(C)$ twice. Are the definitions equivalent? Why do you need both?*

This is not a different definition, equation (6) is a reformulation which is more useful for our purpose. But we indeed define $\vartheta_p$ twice. We now use a new symbol to distinguish $\vartheta_p$ in Equation 7 and 9.

> *Figure 4: "The visualization uses rotationally-symmetric direction fields (known as p-RoSy fields in computer graphics (Vaxman et al., 2016))." I guess that you have used these fields already in Figure 1, so why introduce them only now?*

We have moved this comment to Figure 1.

> *Figure 6: Using a few discrete values cannot illustrate continuity. Also, the "jump" in $\gamma_p$ results from deleting a vertex, so I doubt that this is a fair comparison. Still, I think that it is important to point out to the reader that the value $\gamma_p$ depends on the number of vertices (here, I allow that two edges connected by a vertex are aligned).*

We adjusted the caption to make our point more clear. The last image is a triangle and according to the definition of $\gamma_p$ is, therefore, described by only three vertices. So, it is indeed a fair comparison. The reviewer is right that the value of $\gamma_p$ has a strong dependency of the number of used vertices, this is exactly the point that we are trying to make with this figure. Also, adding vertices artificially to make $\gamma_p$ continuous leads to more problems, as the values for $\gamma_p$ change if we change the number of vertices. But an equilateral triangle should be recognized as an equilateral triangle, no matter if there is an artificial fourth vertex or not. The triangle in our picture and the triangle that the reviewer mentioned (so our triangle with an artificial fourth vertex) both have the shape of an equilateral triangle, yet for one it is $|\gamma_3|$ = 1.0 and for the other one it is $|\gamma_3|$ = 0.935.

While we agree on the reviewers statement about continuity, we did not modify the sentence, as the meaning should be clear.

> *L160: The definition of the center of mass is incorrect as it is not that of an extended object whose contour is defined by a polygon, but only of the set of vertices. In Figure 6 you write "the choice of the center of mass highly influences the value of $\gamma_p$" - is there really a choice of the center of mass? I thought that it was uniquely defined.*

We here only repeat the definition from Armengol-Collado et al. (2023) in order to be able to directly compare our analyses with the results presented therein. We adjusted the caption to be more clear.

> *L166: What is the weighting you refer to in Equation 9?*

We apologize, the reference is to Equation 8. We have modified this.

> *L312: "Quantifying orientational order in biological tissues can be realized by Minkowsky tensors". As mentioned above, you do not really use them, but use Equation (7), which can be defined without reference to Minkowski tensors.*

Eq. (7) is part of the irreducible representations of the Minkowsky tensor. Therefore the sentence is correct.

> *L318: I do not quite understand the link between being able (or not) to compare $q_p$'s for different values of p and the interpretability of $q_2$ and $q_6$. Also, since you introduce $q_p$, how can the question about their comparability be a recurrent challenge? Finally, would you agree that even though a comparison between the absolute values of $q_2$ and $q_6$ is inappropriate, one can still meaningfully compare relative changes as a parameter is changed or when comparing cells in different conditions?*

We have modified the sentence. Furthermore we agree that one can still meaningfully compare relative changes as a parameter is changed, as we do. However, our claim that $q_2$ and $q_6$ are independent, does not allow to conclude any kind of nematic-hexatic phase transition. We have now provided further evidence using the published data of Armengol-Collado et al. (2023), which unequivocally supports this statement. We would also like to remark that the detection of a phase-transition requires a single order parameter, which cannot exist as $q_2$ and $q_6$ are independent.

We have further explained this in the main text.

> *Figure 7: The axes are not labeled.*

We added the labels.

> *L359: "$q_2$ and $q_6$ values cluster tightly", L362 "$q_2$ and $q_6$ values become highly scattered" Please, quantify.*

We kept these formulations but have added statistical measures to these qualitative descriptions, see Supplementary Figures to Fig 7 for the distance correlation and the P-values of the distance correlation. These data support our claim of independence.

> *L362: "each $q_2$ value spans a broad range of $q_6$ values and vice versa, demonstrating their independence". Please, use a quantitative test of statistical independence.*

We have added statistical information by using the distance correlation and statistical tests, see Supplementary Figures to Fig 7. Similar results are obtained for the Pearson correlation and corresponding tests. However, they are not included as the distance correlation is more general.

> *L371: Please, define $Q_2$ and $Q_6$ in the main text.*

We have now added the definition to the Materials and methods section.

> *L420: A reference seems to be missing.*

Thanks for pointing this out. This was a formatting error, we only wanted to cite Balasubramaniam et al. (2021).

> *L425: "strong dependence of cell shape on cell density". But $q_6$ seems to be rather independent of density, see Figure 11. Also, what do you mean by "strong"? Can you quantify?*

The dependency of the cell shape on the cell density is shown in detail in (Eckert et al., 2023). Furthermore, to describe the cell shape the values for all p are needed. So the change in $q_2$ already indicates a change in the overall cell shape even as $q_6$ is barely changing. As we excluded these experimental results now in favor of the experimental data also used in Armengol-Collado et al. (2023), we did not add further evaluations regarding cell density.

> *L453 "These divergences [nonmonotonic dependence of $\gamma_p$ on activity or deformability] highlight the limitations of $\gamma_p$ in capturing consistent patterns". I am not sure to follow your argument here.*

Besides the quantitative differences seen in comparing Fig. 1 and Fig 2 with the corresponding figures in the appendix, these results show qualitative differences. Using a method which is not robust and not continuous leads to qualitative different results. The nonmonotonic dependence of $\gamma_p$ is specific to the method but not to the underlying physics.

> *Appendix 3 - Figure 20: It is not clear how to compare this figure to Figure 3e of Armengol-Collado et al 2023. Please, provide more details.*

Appendix 3 - Figure 20 (Appendix 3 - Figure 25 in the revised version) and Figure 3e in Armengol-Collado et al. (2023) cannot be directly compared. Fig 3e shows results of experiments and multiphase field simulations for one parameter stetting and Fig 20 results of the active vertex model for various parameter settings. But both are considered using $\gamma_p$ and $\Gamma_p$. We have added these computation, see Fig. 13, which indeed reproduces the results from Fig 3e. We refrain from considering corresponding plots to Fig 20 for the multiphase field model, as this first requires computing the vertices and no additional information can be expected.

> ***Reviewer 2:***
>
> *The manuscript lacks statistical information. The following should be addressed: How often have the experiments been performed? How many monolayers have been analyzed? How many time steps have been considered and in what duration? How many cells have been included in the analysis? What are the p-values to determine if $q_p$'s (Figure 2, panel a) and $\gamma_p$'s (Appendix 3-Figure 17, panel a) are significantly different? Same figures: How many cells and experiments have been considered here? Figure 11: What is the density of cells for each condition? Please provide the corresponding values. How significant are the differences? How many times has the experiment been repeated? Figure 12: Due to cell proliferation, the cell density changes over time. Does this need to be taken into account?*

We agree, our information have only been qualitative. We have added the missing information. Especially we added statistical information by using the distance correlation and statistical tests, see Supplementary Figures to Fig. 7. Similar results are obtained for the Pearson correlation and corresponding tests (not included). As we excluded the experimental results previously shown in Figure 11 and Figure 12, in the revised version in favor of the experimental data that is already published in Armengol-Collado et al. (2023), we did not add further statistics regarding this. We added the number of frames and cells in the text.

> *The image analysis part of the Method section states that time-series were xy-drift corrected, and cells were tracked. However, the manuscript does not contain results of dynamical data, timedependent analyses, or discussions of how $q_p$ changes over time. The authors mention that the fluidity of the tissue was confirmed by the MSD, neighbor number variance, and the self-intermediate scattering function, but none of the results are shown in the manuscript. I would like to ask the authors to provide the results and related content in the Method section.*

We have modified the description and removed all parts related to dynamical data. Due to the heavy overload of images in the manuscript we refrain from providing all the results for the phase diagram to distinguish solid and fluid phase. These measures have been provided previously for the considered modeling approaches and provide here only a side remark. Our results do not depend on an exact localization of a solid-fluid phase boundary.

> *Additional information is missing in the Image analysis part of the Method section. Could the authors provide the information on the image analysis steps between obtaining the segmented image and inputting the parameters for the Minkowski tensor? This should include how the normal vectors have been determined and whether this has been done for all pixels along the contour.*

We added further details in the section Extraction of the contour in Experimental setup in Methods and Materials and also provide the code to compute $q_p$ for segmented images.

> *The authors have analyzed low-resolution phase contrast images acquired with a 10x objective to experimentally support their introduced Minkowski tensors. This may have decreased the resolution of the cell boundary detection and its curvature. I strongly suggest imaging the tissue with higher magnification (40x or 63x) and/or fluorescent markers to visualize the cell boundaries in high quality. This would allow the authors to distinguish between circles and circle-like shapes (lines 432-434) and to further investigate differences between MDCK wild-type and MDCK E-cad KO cells.*

We agree that higher resolution of the images would be beneficial. However, we are convinced that this will not influence our findings. Instead of performing the experiments with higher magnification or using fluorescent markers, we have considered the experimental data from Armengol-Collado et al. (2023) to support our results.

> *The authors have coarse-grained the shape function, $\Gamma_p$, and have chosen the active vertex model (Appendix 3-Figure 20) for comparison with the Minkowski tensors, $Q_p$ (Appendix 2 Figure 13). In both figures, the hexatic-nematic crossover does not occur. Armengol-Collado et al. have previously reported that the Voronoi model failed to achieve the hexatic-nematic crossover and argued that this is due to the artificial enhancement of the polygon's hexagonality, leading to high hexatic order at the tissue scale. Since the authors have used the Voronoi-tailing method (line 196), I would like to ask the authors to compare the multiphase field models for $\Gamma_p$ and$Q_p$ instead.*

We would like to mention that we do not consider a Voronoi model but an active vertex model. A Voronoi model is only used for initialization. Both models are certainly related but not identical and claims for a Voronoi model do not need to hold for an active vertex model. The suggested comparison for the multi phasefield model is not an easy task as it requires to compute the vertices from the phase field variables. There are gaps between cells and a reliable algorithm to identify the vertices is a task on its own. We, therefore, refrain from doing these calculations. Instead, we have used the experimental data from Armengol-Collado et al. (2023) for which the polygonal information are provided, see Figure 11. Especially for $p$ = 6, strong differences can be seen by comparing the PDF obtained by the full shape and the polygonal shape. Indeed, the strong hexatic order at the cellular scale is only a consequence of the approximation by polygons. With this result analysing the multi phasefield data by $\gamma_p$ does not add any new information as this first requires an approximation by polygons.

> *The authors show the $q_p$ distributions for the experimental systems (Figure 2, Figure 11). For completeness, I would like to ask the authors to also coarse-grain $q_p$ and $\gamma_p$ of the*

> *experimental data as shown for the computational models in Appendix 2 - Figure 13 and Appendix 2 - Figure 14. It would be interesting to see if the hexatic-nematic crossover appears. I would recommend that the authors avoid using the Voronoi tailing of the experimental system, as this may fail to obtain the crossover as explained in (5) above. Instead, I suggest using the real vertex positions for $\gamma_p$, which can be obtained from the segmented images.*

It remains open what is meant by "the real vertex positions for $\gamma_p$, which can be obtained from the segmented images". Segmenting the images leads to smooth contours, partly even with gaps between cells. As the magnitude of $\gamma_p$ depends on the number of points used in the calculation it is not meaningful to use all points of the contour for calculating $\gamma_p$, as this would lead to artificially low values for $|\gamma_p|$. Identifying the vertex positions for an approximating polygon is an issue of its own and the consequence of this approximation is already mentioned above. For a comparison we therefore added the experimental data from Armengol-Collado et al (2023) and used the provided vertex positions to compute $q_p$ and $\gamma_p$ as well as the raw data and performed the segmentation and used these data to compute $q_p$. See Figure 11. These results confirm our findings and show that the proposed nematic-hexatic phase transition is specific to $\gamma_p$ to characterize shape.

> *In order to show that shape descriptors like the shape function, $\gamma_p$, introduced by Armengol-Collado et al., 'fail to capture the nuance of irregular shapes' (line 445), the authors have compared $\gamma_p$ with the Minkowski tensors, $q_p$, using the same dataset (Figure 1 with Appendix 3 - Figure 16, Figure 2 with Appendix 3 - Figure 17, and Figure 4 with Appendix 3 - Figure 15 Appendix 3). I agree that $\gamma_p$ and $q_p$ are different, not showing identical values. However, I see no evidence in these figures that $q_p$ describes the symmetry of a cell better than $\gamma_p$, since the values are similar and vary quite similarly between different p-atic orders. What is the quantitative difference that shows the failure of the shape function to capture the nuance of irregular shapes?*

The statement already follows from the mathematical properties of robustness and stability, which is illustrated in Fig. 6. The mentioned comparisons for simulation and experimental data only demonstrate that the lack of robustness and stability of $\gamma_p$ also leads to different results if applied to averages of cell measures. The differences are twofold, first the approximation of cells by polygons leads to different results, and second even for polygons different results follow, as only one approach is continuous and the other not. This has strong consequences for the proposed nematic-hexatic phase transition if coarse-grained. Our added results for the experimental data from Armengo-Collado et al. (2023) show that this behavior is not a physical feature but only specific to the use of $\gamma_p$.

> *The authors claim that the Minkowski tensors provide a 'reliable framework' and that this framework 'opens new pathways for understanding the role of orientational symmetries in tissue mechanics and development' (line 78-79). However, the p-atic orders in the experimental systems peak at very low orders of $q_p < 0.3$, which may not allow conclusions about (non-)dominant orientational symmetry(ies) of cells. Can this framework be applied to experimental systems? Since the Minkowski tensors display the independence of the hexatic and nematic symmetry, the variations of cell shapes in experimental systems are too strong to provide any additional results (line 437), as stated by the authors, and no crossover was found, while the crossover was reported by Armengol-Collado et al., what new pathways can be opened to study tissues?*

We have added a comparison with experimental data from Armengol-Collado et al. (2023) and demonstrate that the proposed nematic-hexatic transition is only specific to the use of $\gamma_p$ for characterizing the shape. So our results first of all essentially close the "pathway for understanding the role of orientational symmetries in tissue mechanics and development", which was proposed on this nematic-hexatic transition. On the other side, even if $q_p$ peaks at relatively low values, the results demonstrate independence of the measures for different p's, for two different modeling approaches and two different sets of experimental data. This motivates to consider p-atic order for different p simultaneously. Such theories of "multi"-p-atic liquid crystals, as proposed in the conclusions, are the mentioned new pathways.

> *In principle, the introduced Minkowski tensors integrate the orientation of the normal vectors (Equation 6) and consider the perimeter of the contour (Equation 1). Do the tensors distinguish between convex and concave curvature since both are present in tissues? Does a square with 4 concave and a square with 4 convex edges (same curvature) have the same $q_p$ values?*

For the specific situation of a square with 4 concave or 4 convex edges even p would lead to the same orientation and the same value for $q_p$, as even p have a 180 degree symmetry. Odd p would result in the same value for $q_p$ but in a different orientation $\vartheta_p$. In more general cases, e.g. shapes with concave and convex edges, no general statements can be made. In general the theoretical results on stability of $q_p$ only hold for convex shapes. However, as discussed in Methods and materials the known counterexamples for concave shapes are not relevant for cell shapes.

> *In lines 169-172 and Figure 6, the authors report a jump in $\gamma_p$. Why has the fourth vertex in the last image been removed? The vertices are essential for the calculation of $\gamma_p$. If the fourth vertex is not removed, the following values result: $\gamma_3$ = 0.935 and $\gamma_4$ = 0.474, which leads to changes of the same order of magnitude as those of $q_p$. I think it is therefore not the choice of the center of mass that 'heavily influences the value of $\gamma_p$', but the removal of the fourth vertex.*

We adjusted the caption to make our point more clear. The last image is a triangle and according to the definition of $\gamma_p$ is therefore described by only three vertices. The reviewer is right that the value of $\gamma_p$ has a strong dependency of the number of used vertices, this is exactly the point that we are trying to make with this figure. An equilateral triangle should be recognized as an equilateral triangle, no matter if there is an artificial fourth vertex or not. The triangle in our picture and the triangle that the reviewer described (so our triangle with an artificial fourth vertex) both have the shape of an equilateral triangle, yet for one $|\gamma_3|$ = 1.0 and for the other one it is $|\gamma_3|$ = 0.935. This can be seen even more clearly if even more artificial vertices on the outline of the equilateral triangle are added, which will decrease $|\gamma_3|$ even more. Furthermore, we think there was a misunderstanding regarding our statement about the center of mass. The general problem of $\gamma_p$ - so the dependence of the values on the number of vertices - is independent of the calculation of the center of mass. The exact values of $\gamma_p$ on the other hand depend on the choice of this. We follow Armengol-Collado et al. (2023) and use the mean of all vertex coordinates as center of mass. If the reviewer would use the center of mass of the equilateral triangle and do the same calculations the resulting values for $\gamma_p$ would be different. This is what we meant with 'heavily influences the value of $\gamma_p$'.

> *In Appendix 3 - Figure 18, the authors show that the shape function, $\gamma_6$, exhibits a non-monotonic trend as a function of activity and deformability. I have no objection to this statement. However, I would like to ask the authors to check the values for $\gamma_6$. In the bottom-left corner, for example, $\gamma_6$ = 0.55. This value seems very low to me. In Appendix 3-Figure 20, $|Q_6|$ for R/Rcell = 2 is already in this range, while $|Q_6|$ for R/Rcell = 1 (not shown), corresponding to $\gamma_6$, must be even higher. Also, the parameters $p_6$ = 3.5 and $v_0$ = 0.1 should result in a nearly hexagonal lattice, which should be captured with high $\gamma_6$ values. I would expect $\gamma_6$ to be in the same range as $q_6$.*

Many thanks for pointing this out. There are two different points addressed in this question: The first is if $|\Gamma_\mathrm{p}|$ is too high. We checked the values, $|\Gamma_\mathrm{p}|$ = 0.5075 for $R/R_{cell}$ = 2, so it is lower than $\overline{|\gamma_6|}$ = 0.58. The second question is why $\gamma_p$ and $q_p$ are not in the same value range.

You are right that for a perfectly hexagonal lattice both should give the same value, namely

$\overline{q_6}$ = $\overline{|\gamma_6|}$ = 1.0. However, even at $p_6$ = 3.5 and $v_0$ = 0.1 this is not a perfectly hexagonal lattice anymore and how fast the values of $q_6$ and $|\gamma_6|$ drop if we move away from a perfect hexagon scales differently. As $q_p$ is stable and only changes slightly for slight changes in the shape it makes sense, that $q_p$ is still close to 1.0 . We included an image, see below, of one time step in said parameter to showcase that cells do not form a perfect hexagonal lattice anymore.

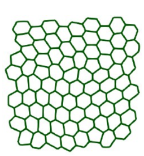
> ***Reviewer 3:***
>
> *Could the authors show why and how this method could bring new information which were missing so far in the understanding of morphogenesis in vitro and in vivo with the current quantification?*

The introduction provides examples of how orientational order and its topological defects can be linked to morphological changes in tissues. The orientational order emerges from the shape of the cells. Most commonly nematic order has been considered, but more recently also hexatic order and even a nematic-hexactic crossover on larger scales. This suggests a mechanical mechanism for morphogenesis, like a phase transition from hexatic to nematic, which would have consequences on the evolution of shape. We demonstrate that the measures $q_2$ and $q_6$ are independent. Furthermore the proposed nematic-hexatic transition is only specific to the use of $\gamma_p$ for characterizing the shape and coarse-graining of the associated order. These measures are not robust and therefore should not be used. Results for the robust measures $q_p$ suggest to consider all $p$ for a coarse-grained theory to model morphological changes in tissues.

> *Could authors show quantitative comparisons between available methods with the same sets of data and highlight pros and cons?*

**Author response image 1.**

Screenshot from $p_6$ = 3.5 and $v_0$ = 0.1



In addition to what was already done for the simulation data we have added data from Armengol-Collado et al. (2023) and compared the results for $q_p$ and $Q_p$ and $\gamma_p$ and $\Gamma_\mathrm{p}$. The theoretical results and the illustrating example in Fig. 6 already show that there are no pros for $\gamma_p$. Other methods belong to the class of bond-order methods and measure neighbor relations instead of shape. We already comment that these methods are inappropriate to classify shape, see Methods and materials, last sentence and Mickel et al. (2013) for a detailed discussion why these methods are not robust.

> *Instead of using phase contrast images, which exhibit curved cell-cell contours, could authors use data with E-cadherin staining instead - as used in many epithelial studies in vitro and in vivo? Could they show both images for wild type and for the E-cadherin KO cell lines with fluorescent readout?*

We are convinced that our results do not depend on the way to visualize the cell contours. Furthermore the images do not provide additional information. To further strengthen the experimental part of the manuscript, we instead analyzed data from Armengol-Collado et al. (2023).

They confirm our findings.

> *The authors acknowledge differences in density between cell lines p. 13 so this calls for new experiments with solid readouts and analysis using comparable experimental conditions.*

Additionally, we analyzed data from Armengol-Collado et al. (2023) which confirm our findings. Our results are now supported by two different modeling approaches and two different experimental settings. Because of redundancy we removed the original experimental data from the revised manuscript.

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